



**STUDENT IDENTITY CHECKING SYSTEM
FOR EXAMINATION ATTENDANCE:
MFU CASE STUDY**

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**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2020

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**A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

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TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
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2020

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ACKNOWLEDGEMENT

This document of the project is a partial fulfillment of the requirement of Senior Project of Mae Fah Luang University. This project is “Student identity checking system for examination attendance : MFU Case Study” is fingerprint scanning technology. That is used to detect fraud in exams and can check information online, website and mobile phone Using the fingerprint scanning module Read each person's fingerprint and use the raspberry pi to write to Control program.

This project would not be successful without the support of the advisors Assoc.Prof.Wg.Cdr.Dr.Tossapon Boongoen, and Our committees, Asst.Prof. Dr. Natthakan Iam-On and Aj. Surapol Vorapatraton, provide good suggestions and comments to improve my project and much additional information.. We would like to thank everyone for their advices and help. We also thank friends in the project group that cooperate as well working in the duties of each person without glitches until this project is successful. Finally, without the kindness of the staff at the School of Information Technology and Registrar of Mae Fah Luang University, this senior project would not have been possible.

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ABSTRACT

Today, technology plays an important role in many people in everyday life and helps to develop the economy even more. It is very important that technology can facilitate. It is what fraudsters or data criminals look at differently in Data traffic for bad use Nowadays, there are many safety technologies that help People can use it safely Therefore, the project organizer realizes the importance

Safety and protection are applied by examining cheating detectors that are caused by the examiner instead and the examiner cannot detect it. The organizer therefore adopted the fingerprint scanning technology Come to use To facilitate supervisors to detect frauds in the exam and can check data online, on websites and mobile, by using the fingerprint scanner module to read each other's fingerprints and use the raspberry pi write to control program.

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

As a result of Mae Fah Luang University has the policy to have an exit examination of the knowledge before graduation. To undergraduate students in all courses, this is to ensure the quality of education for students who will graduate as a university graduate. There are a lot of students for each exam. And to prevent fraud in the form of examinations if the student ID card is forged which makes the examination committee unable to prove the identity of the examinee

This project aims to develop a system for verifying the identity of individuals. By using fingerprints and collecting exam statistics directly in the database. To reduce the errors of the dataset characterization: student ID, name, picture, and fingerprint will be stored in the server. When scanning the fingerprint on the device the information will be sent to compare it with the original database. And when the information is identical, the device screen will display the result of the authentication Note that the project is supported by the student register.

1.2 Objectives

To develop a fingerprint-based student identification system for MFU examinations, where system setting and examination attendance are provided through a web application.

1.3 Scope

1.3.1 The system is to identify student fingerprints to facilitate the verification of the student identity for examination, provided that a database of student details and fingerprint are provided.

1.3.2 Given the consultant and data provided by Division of Registrar at MFU, the system will be evaluated with a sample set of courses and students.

1.3.3 The fingerprint sensor and IoT processing unit can be deployed for any course examination, as an identification number of this whole unit can be mapped

to a specific examination via the web application. Before the examination, a proctor must log-in the web application to accomplish this.

1.3.4 In addition to this system setting, the web application also presents the result of fingerprint matching per student verification and the overall status of attendance.

1.4 Methodology

Within the examination hall, the system is to identify student fingerprints to facilitate the verification of the identity of the student and prevent exam fraud and will alert the results of the fingerprint that scanned the fingerprint has the same information as the database or not and the results of the fingerprint submitted to compare in the original database are reported. When the fingerprints match, the system will report that the data was saved successfully. And when the fingerprint does not match the original database, the system will save the fingerprint for proofreading but it will refuse to save other information by prompting take action again. This system concept that is shown in Figure1.3, can be accomplished through the following stages.

1.4.1 Acquiring system requirements with users, i.e., student register staff, Mae Fah Luang University.

1.4.2 System design including sensor system, web-server monitoring, and database systems.

1.4.3 System implementation for the device.

1.4.4 System evaluation (functionality and user assessment).

1.4.5 Documentation and system manual.

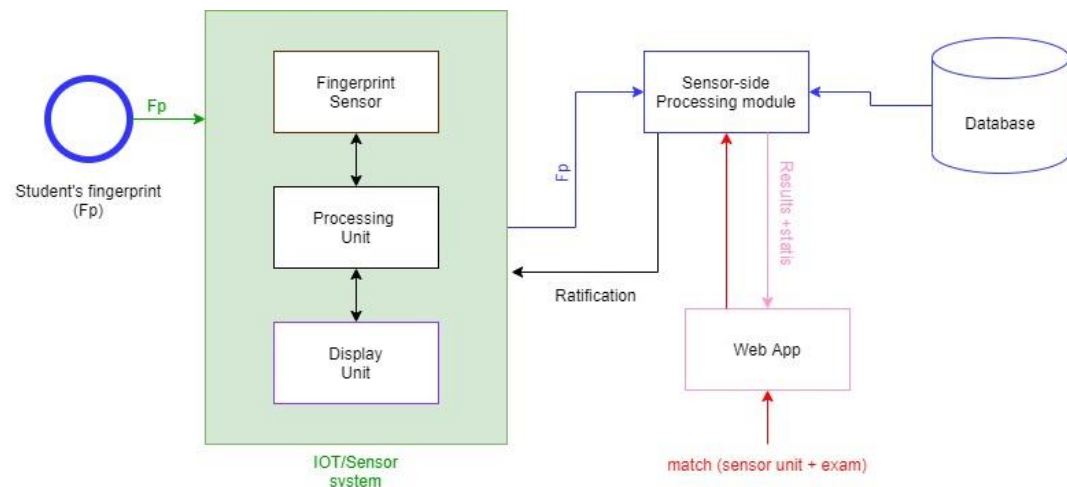


Figure 1.1 Conceptual diagram of the proposed system.

1.5 Planning and Timeframe

This table explains project planning and time frame.

Task/Month	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Finding the topic and the advisor										
Research the information										
Start to make project on software mode										
Write document chapter 1-3										
Prepare for final preproject presentation										
Start to make project on hardware device										
Prepare for final project presentation										

Table 1.1 Working Plan

1.6 Expected Result

The expected result of this project is;

1.6.1 Every part is work done and can work well.

1.6.2 The system can help facilitate staff who come to use the device in the exam room and impress our project.

1.6.3 The system can help staff to find and check student identity.

1.6.4 The system can send a notification to the device and save data on the server.

.

1.7 Equipment, Place and Budget

1.7.1 Equipment

1.7.1.1 Hardware

- Raspberry Pi 4 Model B
- R307 Optical Fingerprint Reader Sensor
- USB 2.0 -TLL UART Serial Converter CP2102
- Jumper Male to Female
- RGB Full Color LED SMD Module KY-009
- SanDisk Ultra MicroSDHC SDCard 32GB Class 10
- Adafruit Pi Protector for Raspberry Pi

1.7.1.2 Software

- Raspbian

1.7.2 Place

Mae Fah Luang University

1.7.3 Budget

The price of equipment below is approximated.

- Raspberry Pi 4 B	1940	Baht
- R307 Optical Fingerprint Reader Sensor	590	Baht
- USB Converter CP2102	70	Baht
- Jumper Male to Female	50	Baht
- RGB Full Color LED SMD	25	Baht
- MicroSDHC SDCard 32GB	380	Baht
- Adafruit Pi Protector for Raspberry Pi	385	Baht

Total 3440 Baht

CHAPTER 2

LITERATURE REVIEW

2.1 Theory

2.1.1 Theory of Biometrics

Biometrics are body measurements and calculations related to human characteristics.[1] Biometrics authentication (or realistic authentication) is used in computer science as a form of identification and access control. It is also used to identify individuals in groups that are under surveillance. Biometric identifiers are the distinctive, measurable characteristics used to label and describe individuals. Biometric identifiers are often categorized as physiological versus behavioral characteristics. Physiological characteristics are related to the shape of the body. Examples include, but are not limited to fingerprint, palm veins, face recognition, DNA, palm print, hand geometry, iris recognition, retina, and odor/scent. Behavioral characteristics are related to the pattern of behavior of a person, including but not limited to typing rhythm, gait, keystroke, signature, behavioral profiling, and voice. Some researchers have coined the term behavior metrics to describe the latter class of biometrics. More traditional means of access control include token-based identification systems, such as a driver's license or passport, and knowledge-based identification systems, such as a password or personal identification number. Since biometric identifiers are unique to individuals, they are more reliable in verifying identity than token and knowledge-based methods; however, the collection of biometric identifiers raises privacy concerns about the ultimate use of this information.

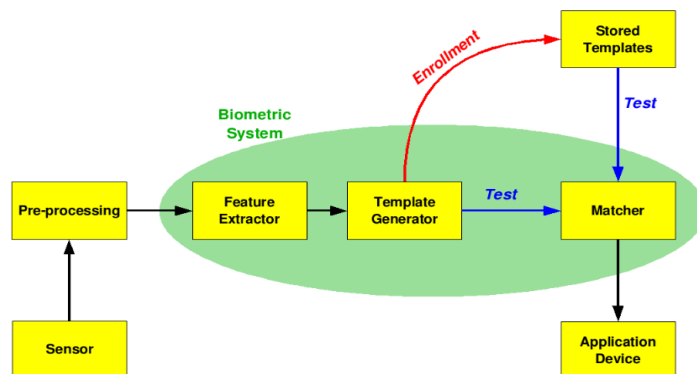


Figure 2.1 Biometrics system

The simple diagram showing the main logical block of a biometric system.

2.1.2 Theory of Fingerprints

A fingerprint is an impression left by the friction ridges of a human finger. The recovery of partial fingerprints from a crime scene is an important method of forensic science. Moisture and grease on a finger result in fingerprints on surfaces such as glass or metal.[2] Deliberate impressions of entire fingerprints can be obtained by ink or other substances transferred from the peaks of friction ridges on the skin to a smooth surface such as paper. Fingerprint records normally contain impressions from the pad on the last joint of fingers and thumbs, though fingerprint cards also typically record portions of lower joint areas of the fingers. Human fingerprints are detailed, nearly unique, difficult to alter, and durable over the life of an individual, making them suitable as long-term markers of human identity. They may be employed by police or other authorities to identify individuals who wish to conceal their identity or to identify people who are incapacitated or deceased and thus unable to identify themselves, as in the aftermath of a natural disaster.

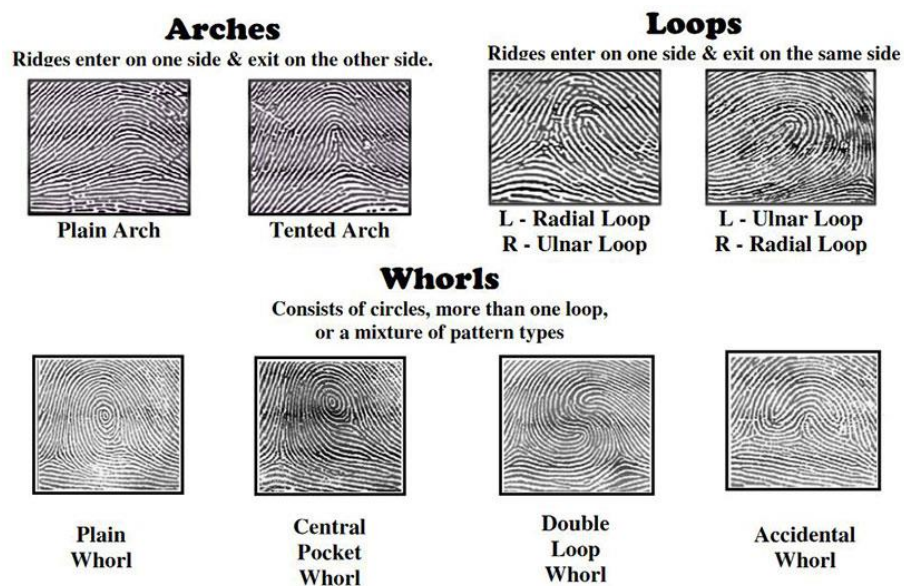


Figure 2.2 Arches, Loops and Whorls of different types

The picture is the basic patterns of human fingerprint.[3]

2.2 Technology and Tool

2.2.1 R307 Optical Fingerprint Reader Sensor



Figure 2.3 Fingerprint Reader Sensor

The Rugged Panel Mount Fingerprint Sensor with Bi-Color LED Ring even has an LED ring built around the detection pad, which can be set to red, blue, or purple (as well as some fading/blinking effects) for a great user experience.[4]

2.2.1.1 Specification Operating

- Supply voltage: 3.6 - 6.0VDC
- Operating current: 120mA max
- Peak current: 150mA max
- Fingerprint imaging time:
- Window area: 14mm x 18mm
- Signature file: 256 bytes
- Template file: 512 bytes
- Storage capacity: 200 templates
- Safety ratings (1-5 low to high safety)
- False Acceptance Rate: <0.001% (Security level 3)
- False Reject Rate: <1.0% (Security level 3)
- Interface: TTL Serial
- Baud rate: 9600, 19200, 28800, 38400, 57600 (default is 57600)
- Working temperature rating: -20C to +50C

- Working humidity: 40%-85% RH
- Full Dimensions: 54 x 20 x 21.5mm

2.2.1.2 Highlights

- Using is simple serial microcontroller, embedded modules can operate this MSP430, 51, AVR, PIC, ARM, FPGA, Raspberry Pi, for Arduino controller can operate this module.
- The module is controlled through the serial port, can use the computer's serial port to control this module.
- Optical fingerprint sensor, high-speed DSP processors.
- High-performance fingerprint matching algorithms.
- Large-capacity FLASH chip hardware and software composition.
- The stable a variety of functions fingerprint registration, fingerprint matching, fingerprint search.

2.2.2 NodeMCU ESP8266

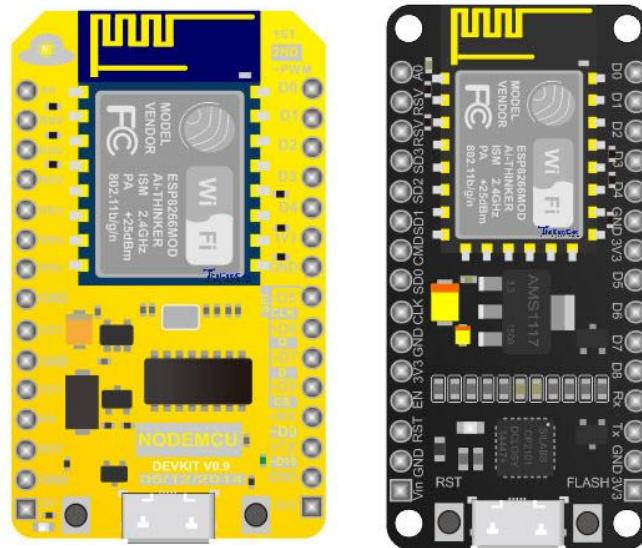


Figure 2.4 ESP8266

ESP8266 offers a complete and self-contained Wi-Fi networking solution,[5] allowing it to either host the application or to offload all Wi-Fi networking functions from another application processor. When ESP8266 hosts the application, and when it is the only application processor in the device, it is able to boot up

directly from an external flash. It has integrated cache to improve the performance of the system in such applications, and to minimize the memory requirements. Alternately, serving as a Wi-Fi adapter, wireless internet access can be added to any microcontroller-based design with simple connectivity through UART interface or the CPU AHB bridge interface.

2.2.2.1 ESP8266 Features

- 802.11 b/g/n protocol
- Wi-Fi Direct (P2P), soft-AP
- Integrated TCP/IP protocol stack
- Integrated TR switch, balun, LNA, power amplifier and matching network
- Integrated PLL, regulators, and power management units
- +19.5dBm output power in 802.11b mode
- Integrated temperature sensor
- Supports antenna diversity
- Power down leakage current of $< 10\mu\text{A}$
- Integrated low power 32-bit CPU could be used as application processor
- SDIO 2.0, SPI, UART
- STBC, 1×1 MIMO, 2×1 MIMO
- A-MPDU & A-MSDU aggregation & 0.4μs guard interval
- Wake up and transmit packets in $< 2\text{ms}$
- Standby power consumption of $< 1.0\text{mW}$ (DTIM3)

2.2.3 Apache XAMPP



Figure 2.5 Apache XAMPP program

XAMPP is the most popular PHP development environment
XAMPP is a completely free,[6] easy to install Apache distribution

containing MySQL, PHP, and Perl. The XAMPP opensource package has been set up to be incredibly easy to install and to use.

2.2.3.1 Apache XAMPP Features

- Access Controls, Permissions
- Backup and Recovery
- Customizable Templates
- Data Migration
- Data Quality Control
- Data Replication
- Database Conversion
- Document Indexing
- E-forms
- Full Text Search
- Image Editor
- In-Memory Processing
- Match & Merge
- Rich Text Editor
- SEO Management
- Version Control

2.2.4 Visual Studio



Figure 2.6 Visual Studio program

Visual Studio Code is a lightweight[7] but powerful source code editor which runs on your desktop and is available for Windows, macOS and Linux. It comes with built-in support for JavaScript,

TypeScript and Node.js and has a rich ecosystem of extensions for other languages (such as C++, C#, Java, Python, PHP, Go) and runtimes (such as .NET and Unity). Begin your journey with VS Code with these introductory videos.

2.2.5 Arduino IDE



Figure 2.7 Arduino IDE program [8]

2.3 Relate Work

2.3.1 Fingerprint Sensor Based Self Bike Starter

Interfacing Fingerprint Sensor with Arduino to design Fingerprint Sensor Based Self Bike Starter using Arduino.[9] The type of fingerprint module we are using is the R305 Fingerprint Scanner Module. Security is a major concern in our day today life, and digital locks have become an important part of these security systems. Fingerprint sensor-based is one of the safest bike starting systems as it has the ability to identify and distinguish every person individually without making any error. Also, the module is very small that it can be kept anywhere, and with the portability feature and less power consumption, you can carry it to any place as well.



Figure 2.8 Fingerprint Sensor Based Self Bike Starter

2.3.2 Fingerprint Sensor Based Door Lock Security System

This project is a fine combination of “Biometrics technology” and “Embedded system technology”.[10] We have seen a Password-based security system and RFID based security system, which is insecure and anyone can have access. The main feature or specialty of the fingerprint is that it is unique. It gives a high level of security than other security systems. So in this project, we will learn how to make Fingerprint Sensor Based Door Lock Security System. We will simply interface the Fingerprint Sensor Module with the Development Board. We will use a Solenoid valve to lock/unlock the door. The display will display the status of matched/unmatched finger and door locking and unlocking status.

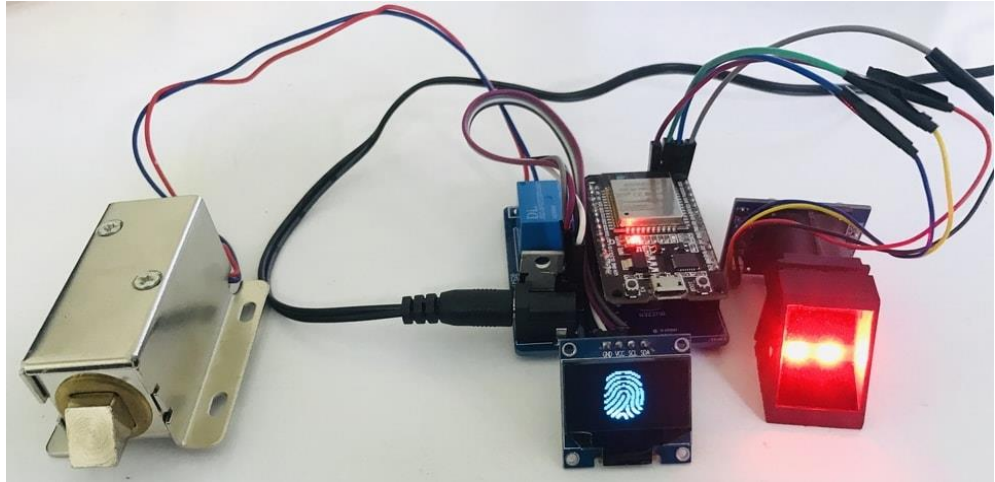


Figure 2.9 Fingerprint Sensor Based Door Lock Security System

2.4 Comparison between related work and the proposed project

This table compares the functions and system of the fingerprint sensor for each project.

Project	Raspberry Pi	Show Alert	Collect data	Connect to the server	Communicate with User
Fingerprint Sensor Based Self Bike Starter		/	/		
Fingerprint Sensor Based Door Lock Security System		/	/		/
STUDENT IDENTITY CHECKING SYSTEM FOR EXAMINATION ATTENDANCE: MFU CASE STUDY		/	/	/	/

Table 2.1 Compare Project

CHAPTER 3

ANALYSIS AND DESIGN

3.1 System Functionalities

This section is to visualise the functionalities provided by the proposed system, in terms of the overall system functionality that is shown by the system diagram, the top-view flowchart that explains processing main stages, and the use case analysis shown with the use case diagram, respectively.

3.1.1 System Diagram

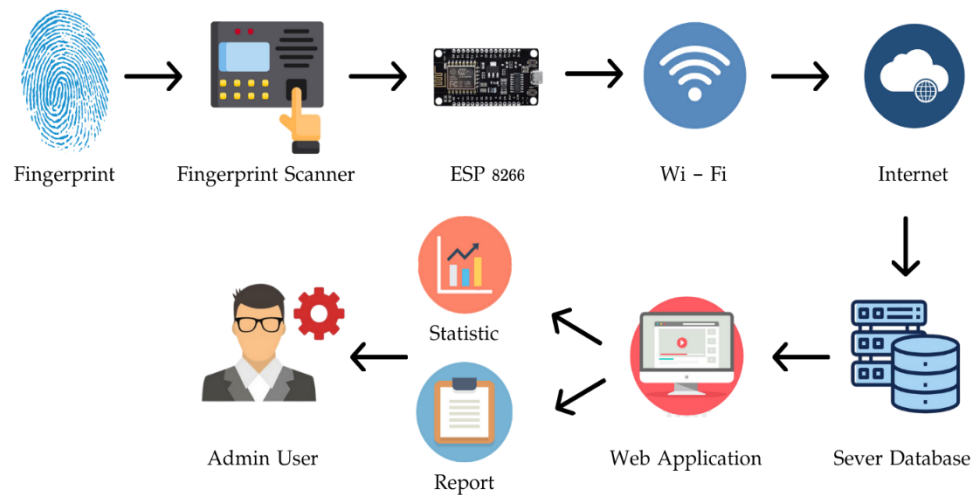


Figure 3.1 System Diagram

From picture 3.1 Start by placing a finger on the fingerprint scanner. Then the machine will scan and send the data to the server to compare the fingerprints in the database and will show id, name, serial number, fingerprint id, date, time in, time out, and show the text “Welcome student name” on OLED.

3.1.2 Top-view Flowchart

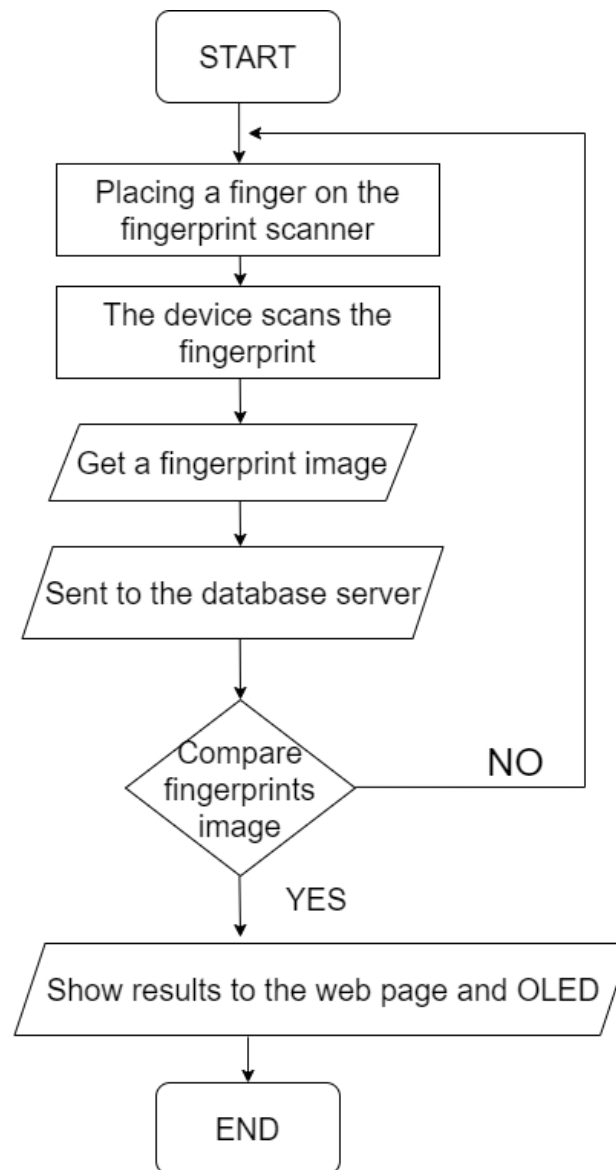


Figure 3.2 Top-view Flowchart

From Figure 3.2 this flowchart shows the operation of the student identification system when entering the examination room. From the scanning process, the fingerprint is compared and shows the result of all processing.

3.1.1.1 Fingerprint processing

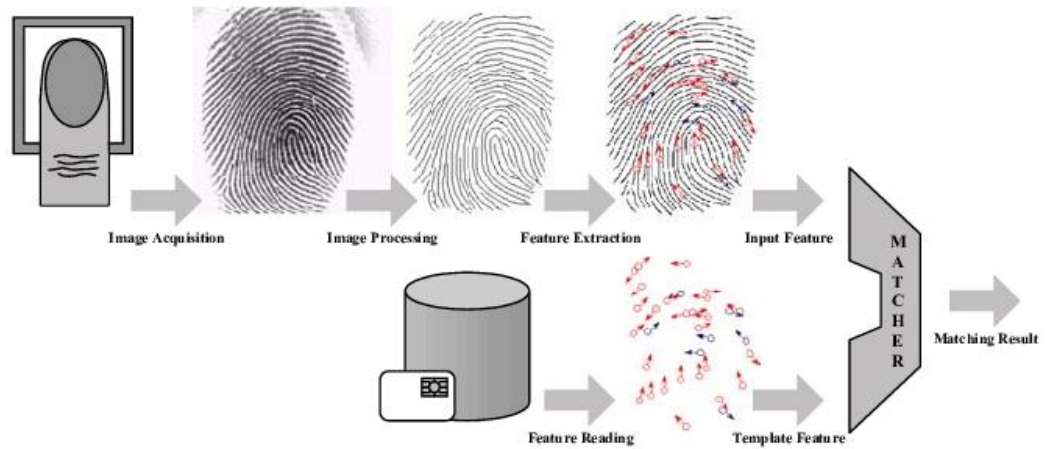


Figure 3.3 Fingerprint processing

Tasks involved in the authentication process: fingerprint image acquisition, image processing and feature extraction stages for the query fingerprint;[11] fingerprint feature reading for the template fingerprint; and fingerprint matching of both, template and query fingerprints.

3.1.1.2 Website processing

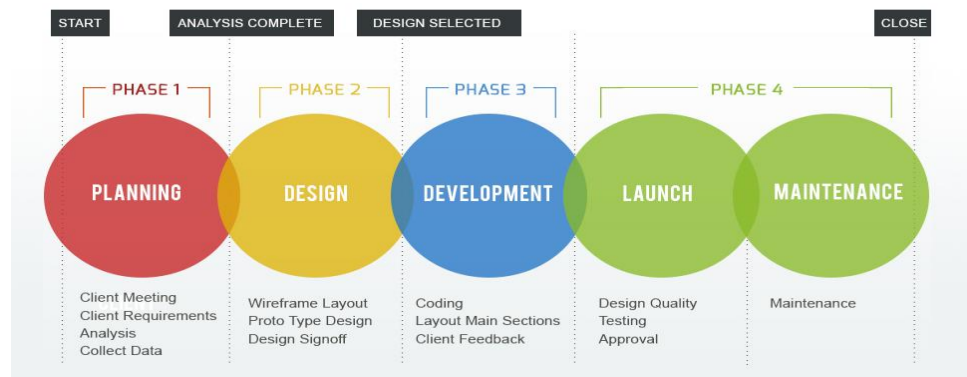


Figure 3.4 Website processing[12]

Well-designed websites offer much more than just aesthetics.[16] They attract visitors and help people understand the product, company, and branding through a

variety of indicators, encompassing visuals, text, and interactions. That means every element of your site needs to work towards a defined goal.

3.1.1.3 Database processing

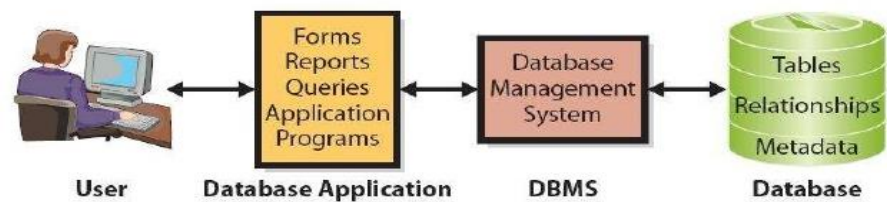


Figure 3.5 Database processing

The purpose of a database is keeping track of things that involve more than one theme.[13]

3.1.2 Use-Case Diagram

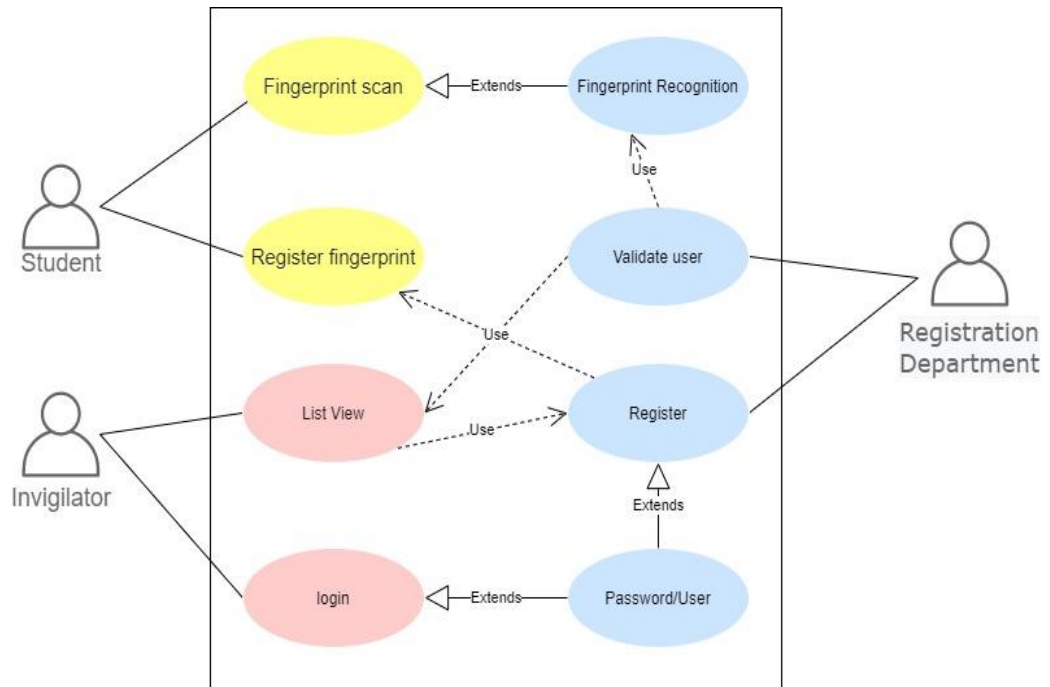
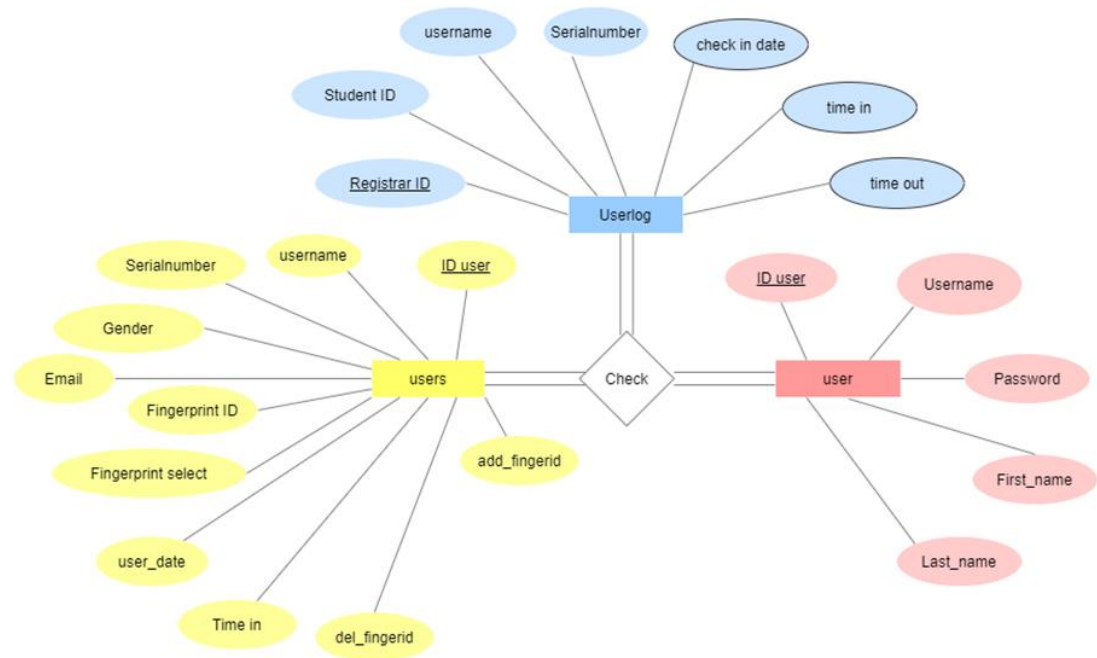


Figure 3.6 Use-Case Diagram

From figure 3.6 this use case diagram can explain describe the scope of access for each type of user.

3.2 Data design

3.2.1 ER Diagram



biometricattendace users_logs
id : int(11)
username : varchar(100)
serialnumber : double
fingerprint_id : int(5)
checkindate : date
timein : time
timeout : time

biometricattendace users
id : int(11)
username : varchar(100)
serialnumber : double
gender : varchar(10)
email : varchar(50)
fingerprint_id : int(11)
fingerprint_select : tinyint(1)
user_date : date
time_in : time
del_fingerid : tinyint(1)
add_fingerid : tinyint(1)

biometricattendace user
ID : int(5)
Username : varchar(50)
Password : varchar(200)
Firstname : varchar(100)
Lastname : varchar(100)
Userlevel : varchar(1)

Figure 3.7 ER Diagram

ER Diagram is a model used to describe the structure of a database, written as a picture. Explaining the structure and relationship of information includes Entity, Attribute, and Relationship.

3.2.2 Data dictionary

A data dictionary, or metadata repository, as defined in the IBM Dictionary of Computing, is a "centralized repository of information about data such as meaning, relationships to other data, origin, usage, and format". Oracle defines it as a collection of tables with metadata. The term can have one of several closely related meanings pertaining to databases and database management systems (DBMS) is a document describing a database or collection of databases, an integral component of a DBMS that is required to determine its structure, and the piece of middleware that extends or supplants the native data dictionary of a DBMS. (Ref: https://www.en.wikipedia.org/wiki/Data_dictionary)

Attributes Name	Description	Data Type	Data Size	Key Type	Reference Table
ID	ID	Int	11	PK	
Username	Username	Varchar	100		
Serialnumber	Serialnumber	Varchar			
Fingerprint_id	Fingerprint_id	Varchar	5		
Checkindate	Checkindate	Date			
Timein	Time				
Timeout	Time				

Table3.1 User data dictionary

Attributes Name	Description	Data Type	Data Size	Key Type	Reference Table
ID	ID	int	11	PK	
username	username	Varchar	100		
serialnumber	Student ID	Bouble			
gender	gender	Varchar	50		
email	email	Varchar	100		
fingerprint_id	fingerprint_id	Int	11		
fingerprint_select	fingerprint_select	Tinyint	1		
user_date	user_date	Date			
time_in	time_in	Time			
del_fingerid	del_fingerid	Tinyint	1		
add_fingerid	add_fingerid	Tinyint	1		

Table 3.2 Users data dictionary

Attributes Name	Description	Data Type	Data Size	Key Type	Reference Table
ID	ID	Int	5	PK	
Username	First Name	Varchar	50		
Password	Password	Varchar	200		
Firstname	Firstname	Char	100		
Lastname	Lastname	Varchar	100		
Userlevel	Userlevel	Varchar	1		

Table 3.3 Users Login data dictionary

3.3 UI design

3.3.1 Site Structure

Site Structure or site structure is how the site is up or how each web link pages together.[14] If any website is well organized, it will help the Search Engine Crawler understand and find things. Within the website faster.

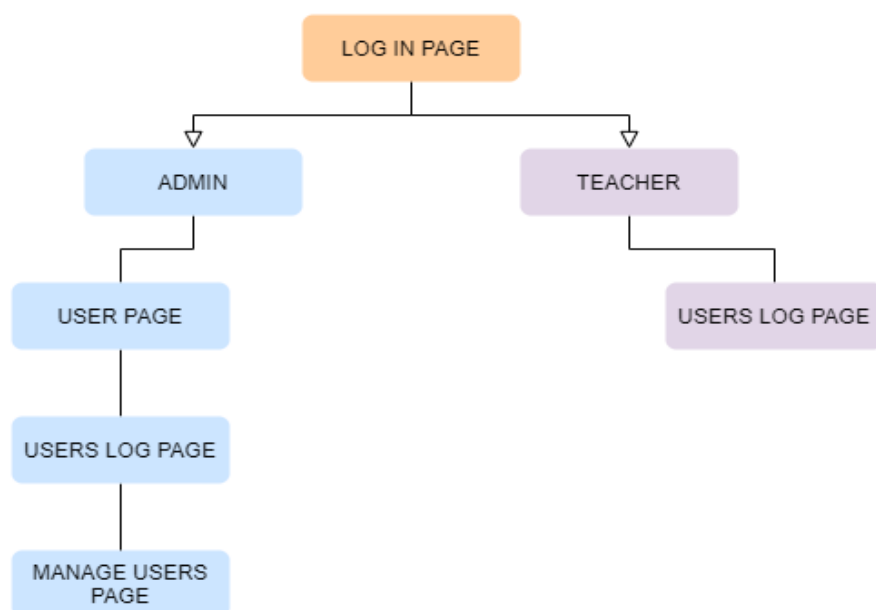


Figure 3.8 Site Structure

3.3.2 Wire frame for Invigilator

Wireframe is the layout or layout of the website. To give an overview of what the website looks like It focuses on the structure,[15] elements, details and continuity on the web page such as text, images, buttons and where each component should be placed. But not yet designed to be beautiful

Wireframe creation is very useful for web design. It is the part that helps to show clear ideas before designing beautiful web pages. It also gives you a glimpse of the web pages or pages that can show potential problems. And fix that problem before proceeding to make a beautiful design.

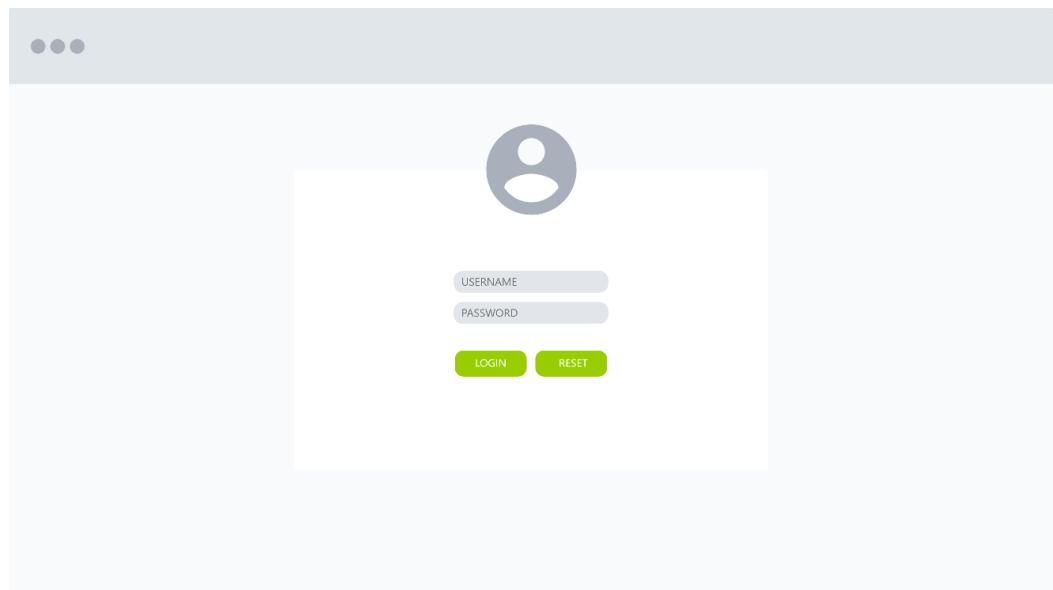


Figure 3.9 Login Page (Admin)

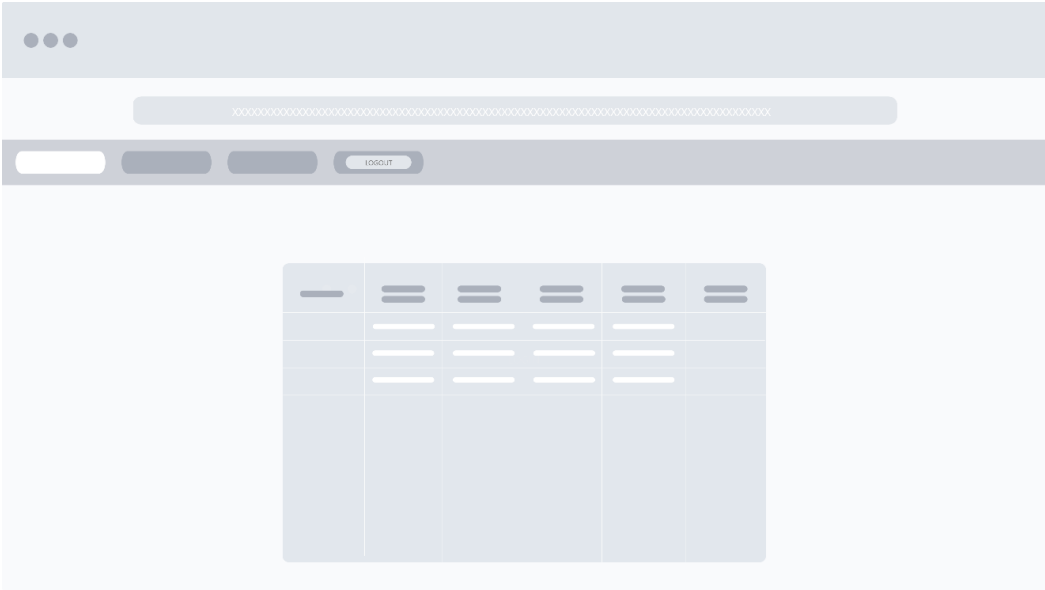


Figure 3.10 Users Page (Admin)

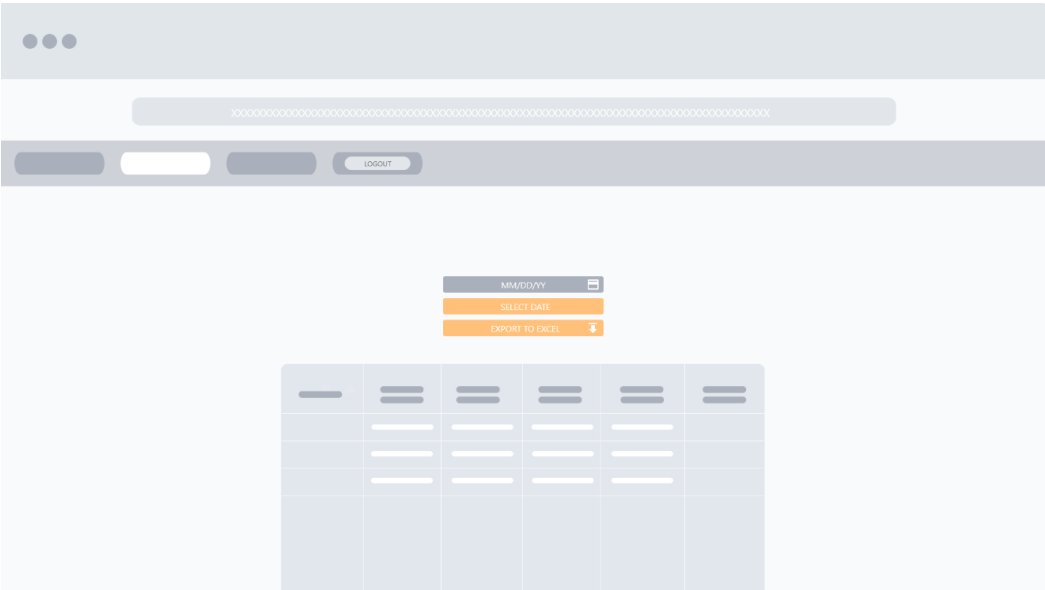


Figure 3.11 Users login page (Admin)

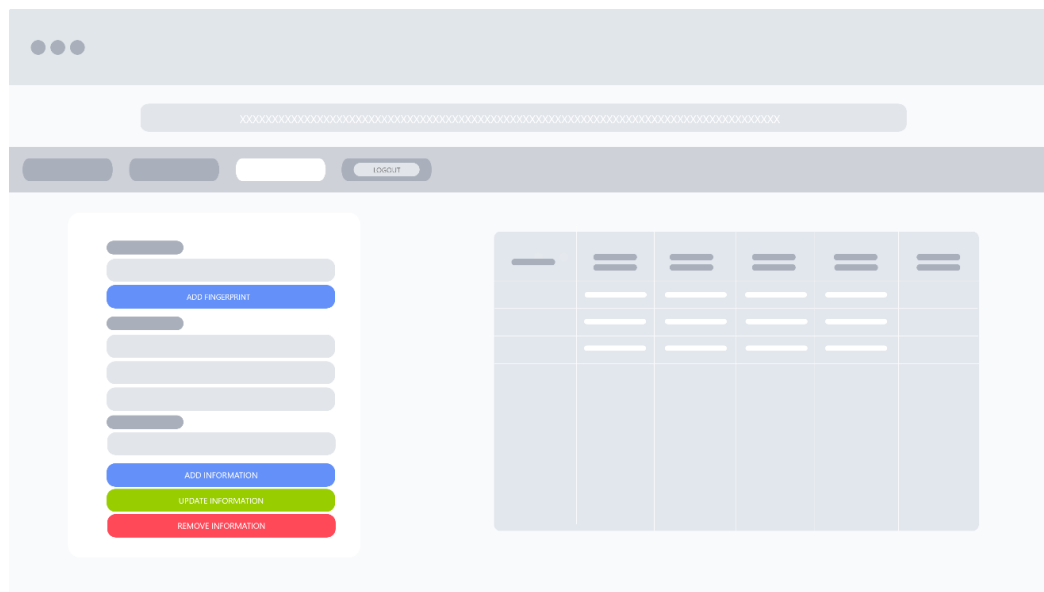


Figure 3.12 Manage users page (Admin)

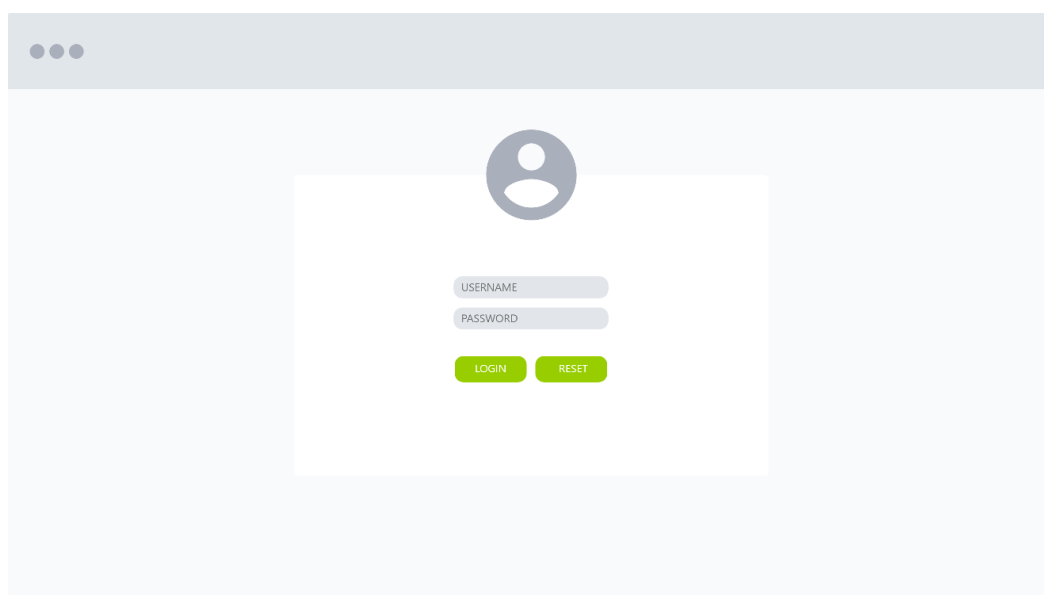


Figure 3.13 Login page (Teacher)

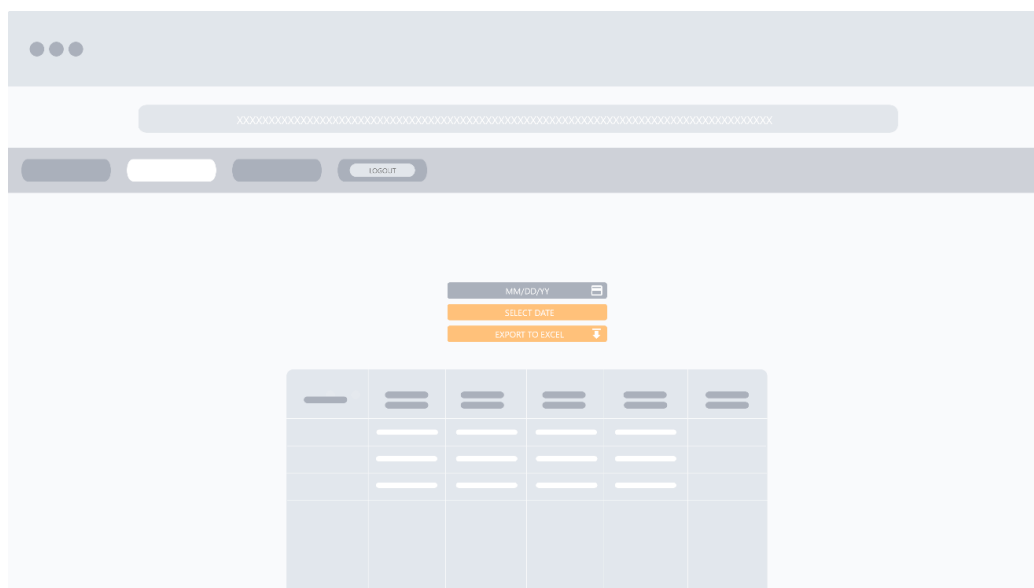


Figure 3.14 Users login page (Teacher)

3.4 Hardware design

3.4.1 Fingerprint Reader Sensor

Connect R307 Optical Fingerprint Reader Sensor, Jumper wire, ESP 8266 and USB type B together, the Vcc pin from the sensor is connected to the 3.3V power supply pin and GND pin to the power side GND pin of the ESP 8266 board. The TX pin is connected to pin D6 and pin RX is connected to pin D5 on the digital side of the ESP 8266 board.

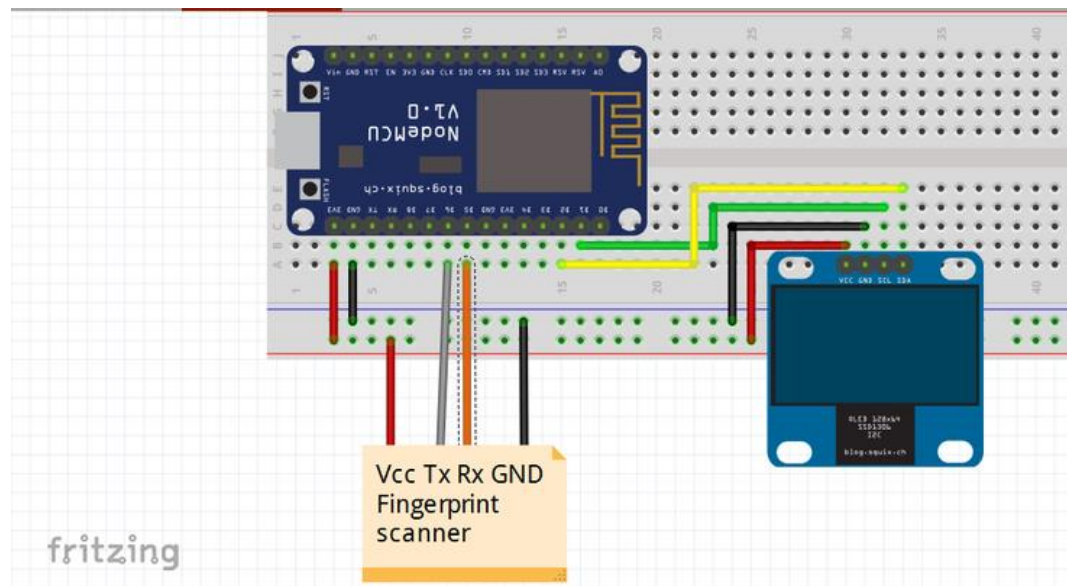


Figure 3.15 Diagram for connecting finger print circuit

CHAPTER 4

RESULTS

In this chapter, we will mention about the result of Student identity checking system for examination attendance. By scanning the fingerprint through the R307 Optical Fingerprint Reader Sensor are send to the database server for comparing with data and will display on website.

4.1 Work procedure

4.1.1 Connecting all devices

4.1.1.1 Connect R307 Optical Fingerprint Reader Sensor with ESP 8266 and OLED

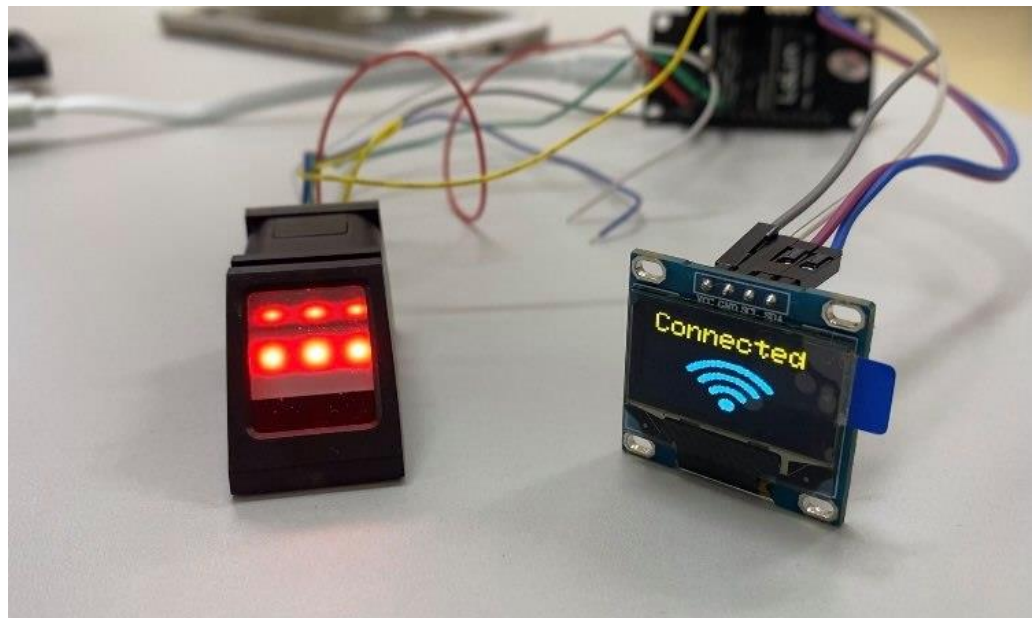


Figure 4.1 Connecting all devices

4.1.1.2 Install Library (Library) to run R307 Optical Fingerprint Reader Sensor by pressing Clone or download can be installed at <https://github.com/adafruit/Adafruit-Fingerprint-Sensor-Library>

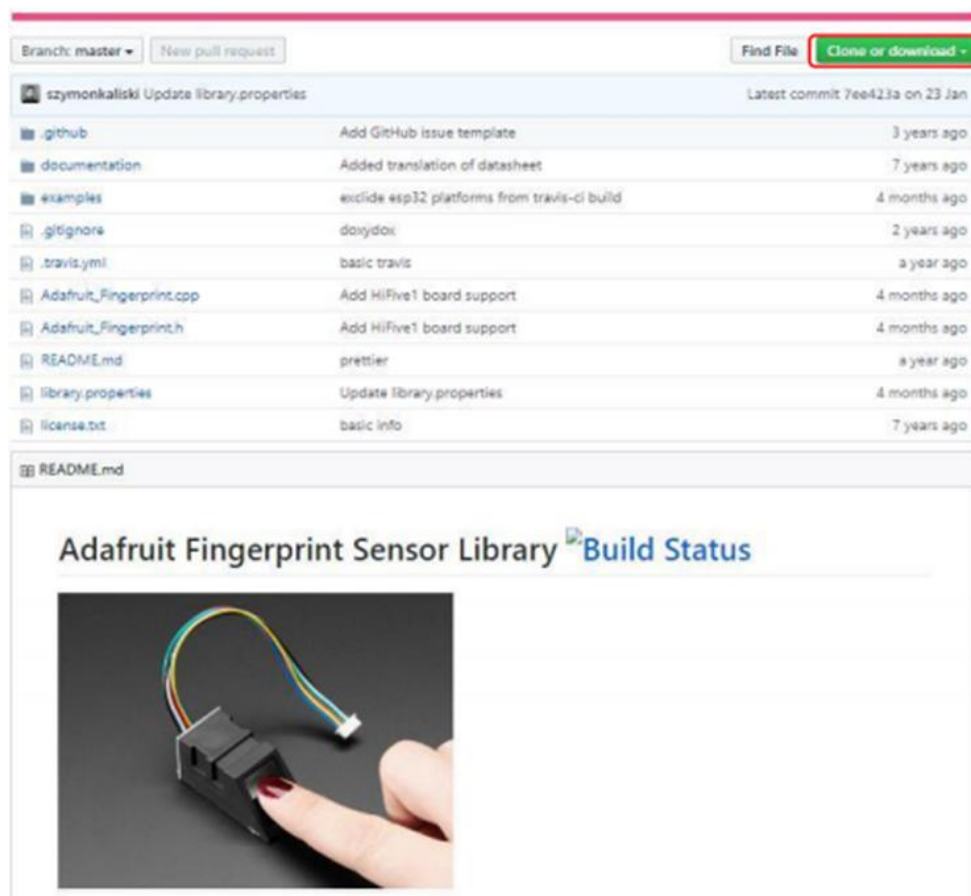


Figure 4.2 Install Library adafruit fingerprint

4.1.1.3 After Follow the step 2. Extract the files and place them in the libraries of the Arduino IDE.

4.1.1.4 Connect Micro USB cable to Computer, open Arduino IDE, then open Code by pressing file -> example -> Adafruit Fingerprint Sensor Libraries, then will receive Code for registration and Upload Code for registration respectively. Next, enroll the fingerprints of 5 people to verify the length and accuracy of identifying images.

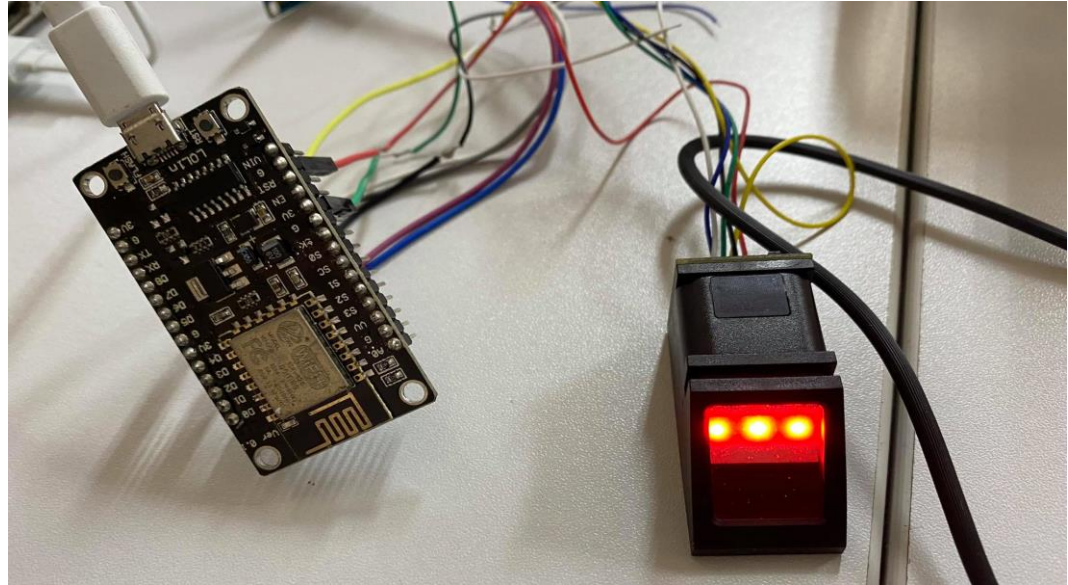


Figure 4.3 Result when we connect the fingerprint circuit

4.1.2 OLED screen

In the experiment, the OLED screen will be experimented by connecting the OLED with the ESP 8266 to be the display side (Output) that while working, now working in which part or in what stage is working.

4.1.2.1 In front of OLED I2C, ESP 8266, Jumper wire, and Micro USB interface, ESP 8266 Vcc pin is connected to 3.3V power supply pin and GND pin to power side GND pin and SDA pin is connected to SDA pin And pin SCL connected to the digital side SCL pin of the ESP 8266 board.

4.1.2.2 Install Library (Library) to run OLED I2C of the OLED screen by going to <https://github.com/fdebrabander/Arduino-LiquidCrystalI2C-library> After that press Clone or download.

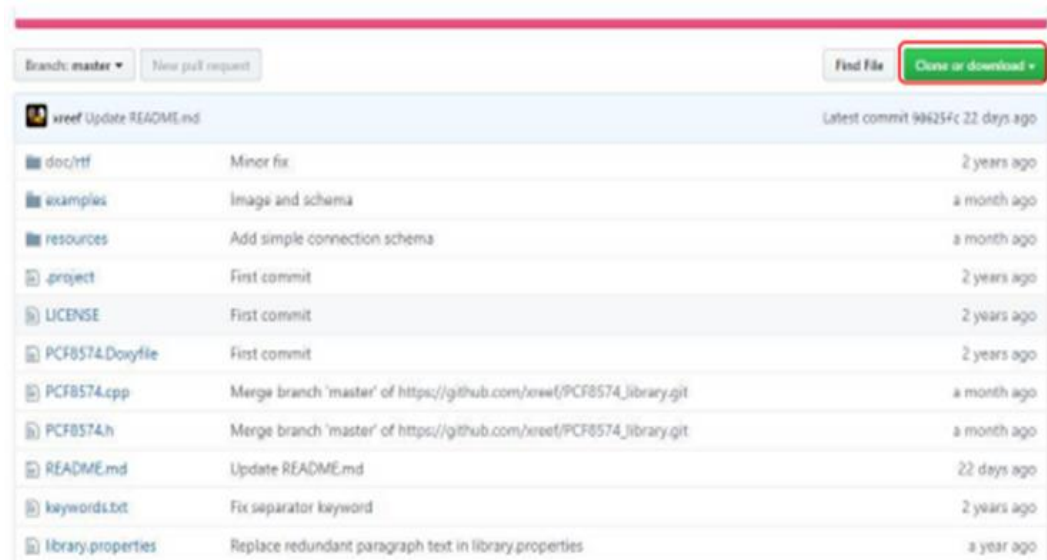


Figure 4.4 Install Library OLED

4.1.2.3. Extract the files and place them in the library of the Arduino IDE.

4.1.2.4. Connect the Micro USB cable to Computer, then open the code in the image by pressing file -> example -> LiquidCrystal -> HelloWorld and uploading the code for testing the screen into the ESP 8266 board via the Arduino IDE

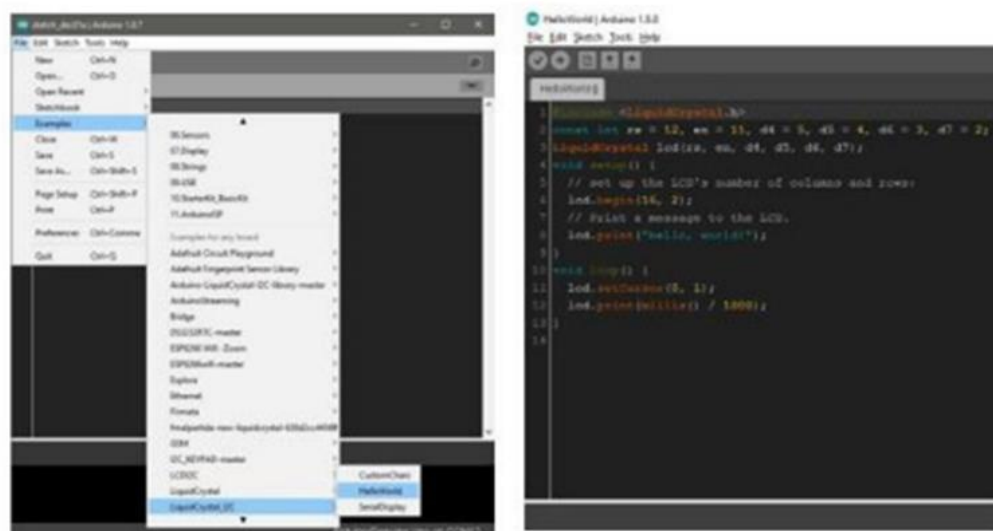


Figure 4.5 Upload code for Testing Screen OLED

4.1.2.5 Start-up test

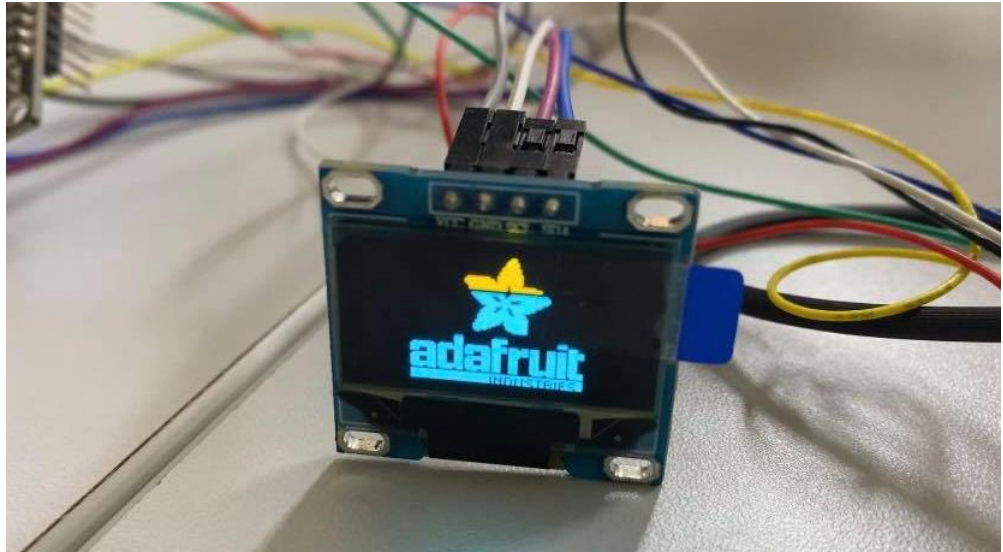


Figure 4.6 OLED screen when start-up

4.1.2.6 esp 8266 is connected to wifi



Figure 4.7 OLED screen show to connect WI-FI

4.1.2.7 Scan the fingerprint to register

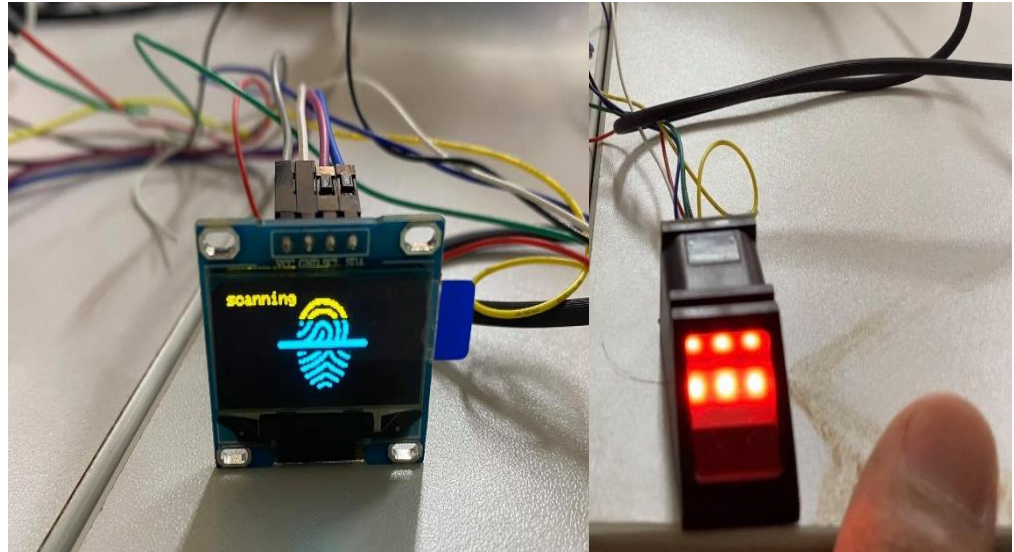


Figure 4.8 OLED screen show to register finger print

4.1.2.8 Scan for check name

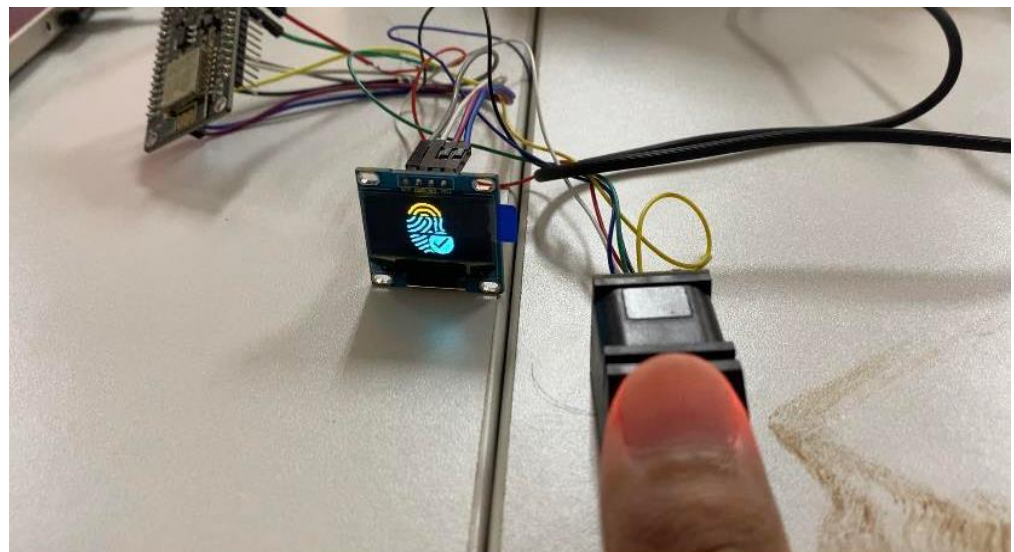


Figure 4.9 OLED screen when scanning a fingerprint to check the name

4.1.2.9 When the scan is not success and if the person has not registered

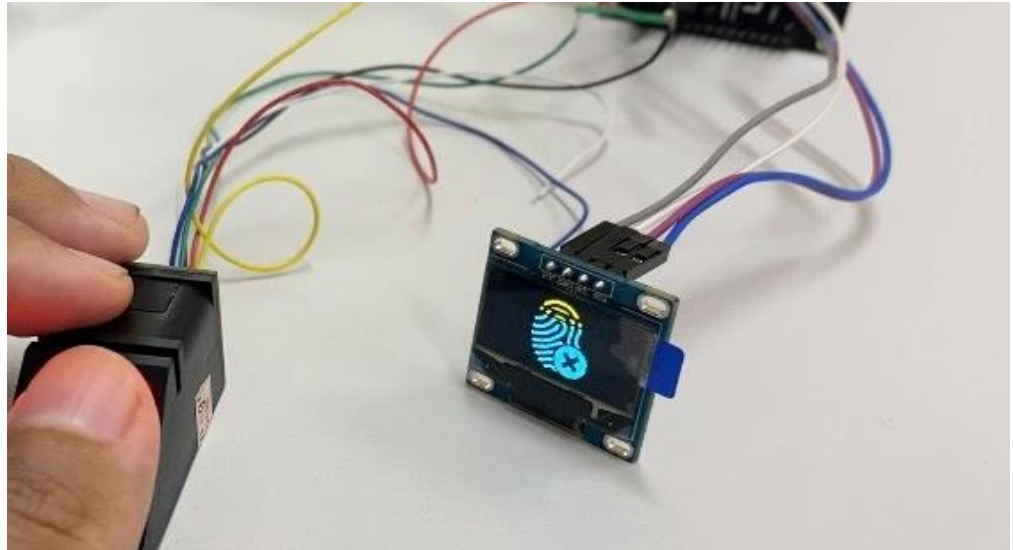


Figure 4.10 OLED screen shows a screen when scanning is unsuccessful

4.1.3 Wireless communication

4.1.3.1 Connect all devices to NodeMCU.V1.0 by connecting NodeMCU.V1.0 Vcc to 3.3V power supply pin and GND pin to power side GND pin, TX pin to pin D6 and RX pin. Connect to pin D5 on the digital side of the ESP 8266 board.

4.1.3.2 Open the Arduino IDE and select File -> Preferences and enter the file address. http://arduino.esp8266.com/stable/package_esp8266com_index.Json

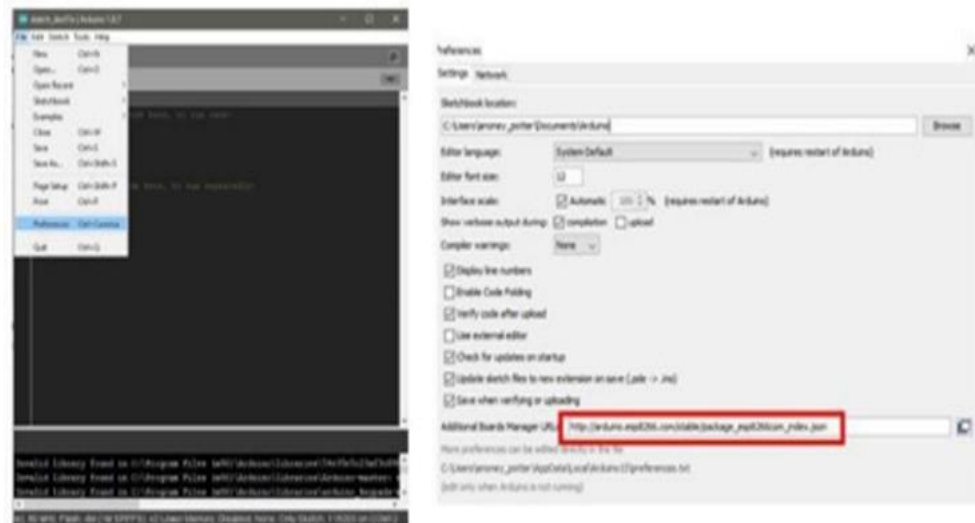


Figure 4.11 Preferences esp8266 file address.

4.1.3.3. After completing step 2, select Sketch -> Include Library and Manage libraries. Respectively and enter esp8266.

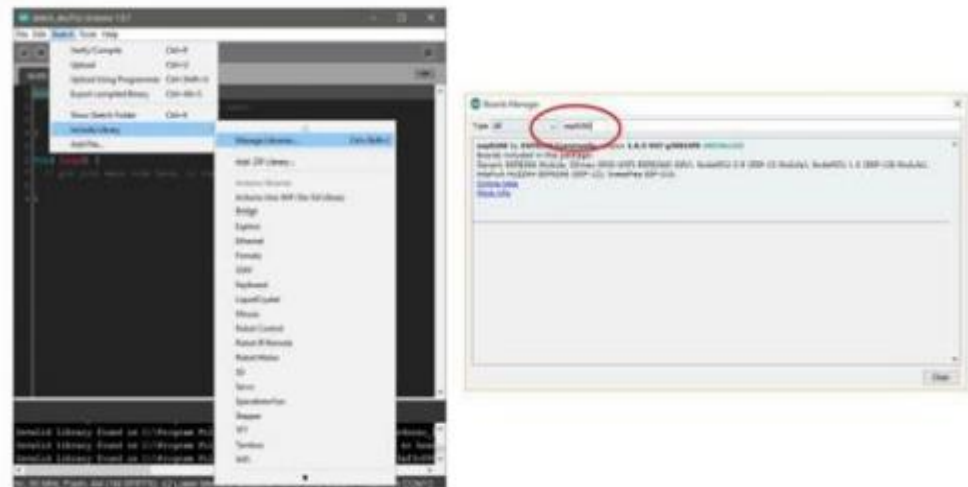


Figure 4.12 Manage Library esp 8266

4.1.3.4 Connect the wireless communication experiment device to the computer and upload the code in the written place for simulating wireless communication test

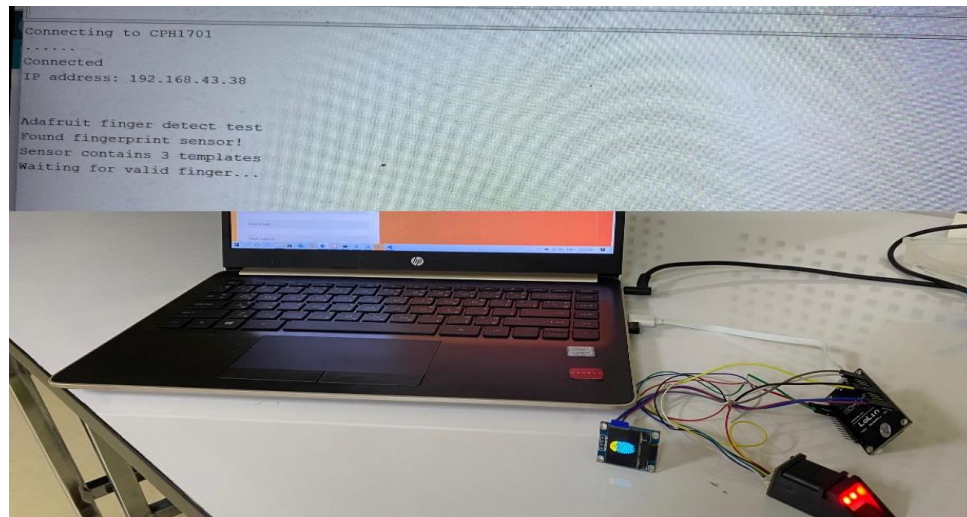
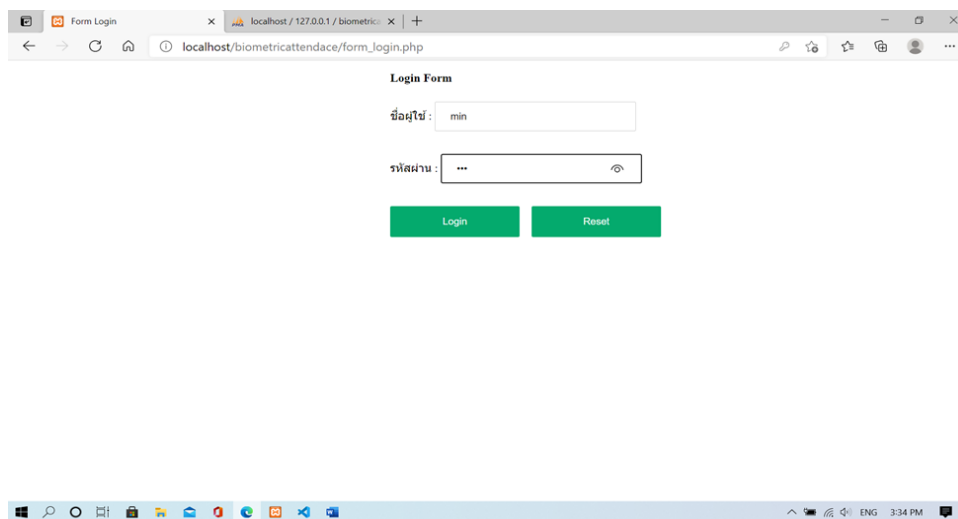


Figure 4.13 Result of Wireless communication

4.1.4 Web page

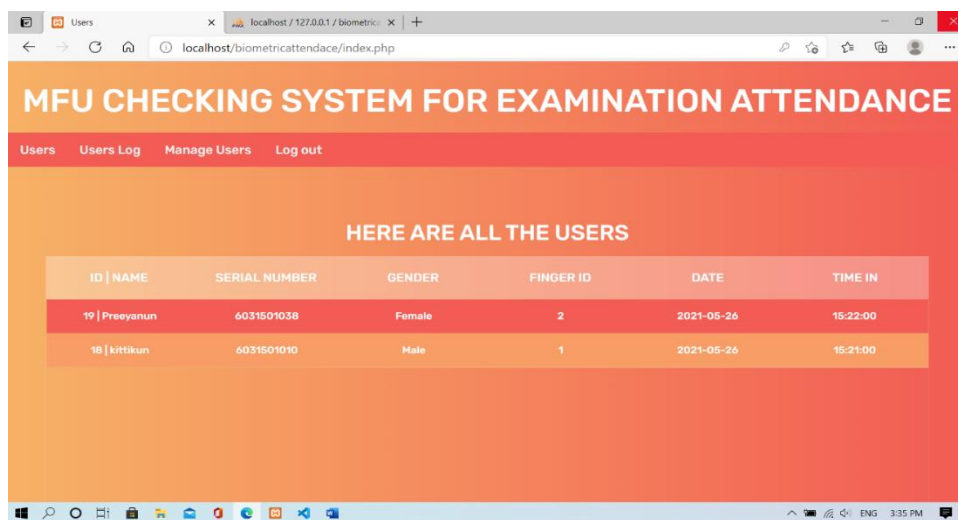
4.1.4.1 Login to the admin webpage



The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/form_login.php'. The page title is 'Form Login'. It contains a login form with two input fields: 'Username' with the value 'min' and 'Password' with a masked value '***'. Below the fields are two green buttons labeled 'Login' and 'Reset'.

Figure 4.14 Admin login page

4.1.4.2 Page showing all user registered information



The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/index.php'. The page title is 'Users'. The main heading is 'MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE'. Below the heading is a navigation bar with links: 'Users', 'Users Log', 'Manage Users', and 'Log out'. The main content area is titled 'HERE ARE ALL THE USERS' and contains a table with the following data:

ID	NAME	SERIAL NUMBER	GENDER	FINGER ID	DATE	TIME IN
19	Preeyanun	6031501038	Female	2	2021-05-26	15:22:00
18	kittikun	6031501010	Male	1	2021-05-26	15:21:00

Figure 4.15 Users page (Admin)
Shows all the registered users

4.1.4.3 Register fingerprint by specifying the fingerprint ID that you want to register

Figure 4.16 Manage users page 1
Set the fingerprint ID that you want to register.

Figure 4.17 Manage users page 2
After scanning the fingerprint that you want to register

4.1.4.4 Add information by filling in the fields.

The screenshot shows the 'Manage Users' page with two main sections: 'User Fingerprint ID' and 'User Info'.

User Fingerprint ID: A form with a label 'Enter Fingerprint ID between 1 & 127:', a text input field, and an 'Add Fingerprint ID' button.

User Info: A form with a dropdown menu for time selection (showing 03:37 PM) and radio buttons for 'Female' and 'Male'.

Table of Existing Users:

FINGER ID	NAME	GENDER	S.NO	DATE	TIME IN
3	vivo	Male	60315010	2021-05-26	15:37:00
2	Preeyanun	Female	6031501038	2021-05-26	15:22:00
1	kittikun	Male	6031501010	2021-05-26	15:21:00

Figure 4.18 Manage users page 3
Fill out all the information in the fields to register

4.1.4.5 Press Add Fingerprint

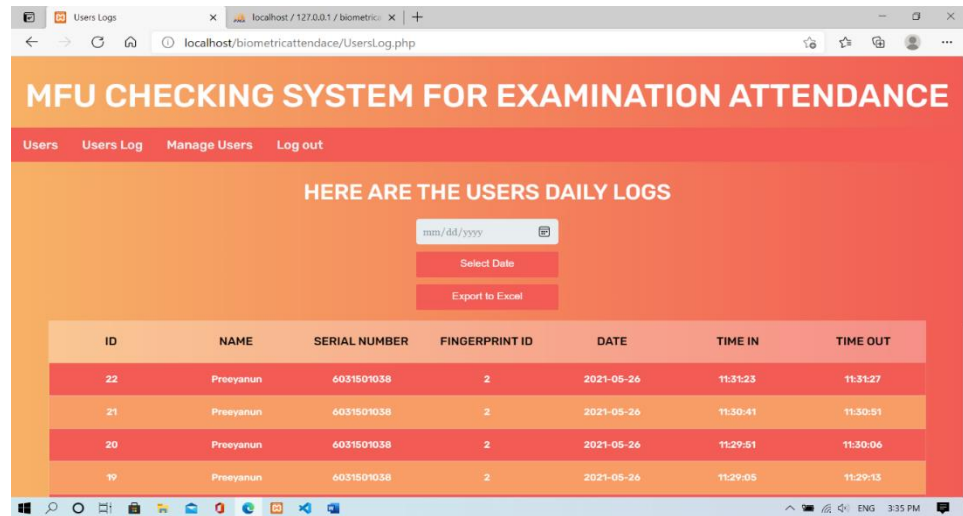
The screenshot shows the 'Manage Users' page after registration. The 'User Fingerprint ID' and 'User Info' forms are still visible. The table of existing users is updated with the new user entry.

Table of Existing Users:

FINGER ID	NAME	GENDER	S.NO	DATE	TIME IN
3	vivo	Male	60315010	2021-05-26	15:37:00
2	Preeyanun	Female	6031501038	2021-05-26	15:22:00
1	kittikun	Male	6031501010	2021-05-26	15:21:00

Figure 4.19 Manage users page 4
After completing the registration

4.1.4.6 Showing real-time information when fingerprint scanning occurs. To check the name



MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE

Users Users Log Manage Users Log out

HERE ARE THE USERS DAILY LOGS

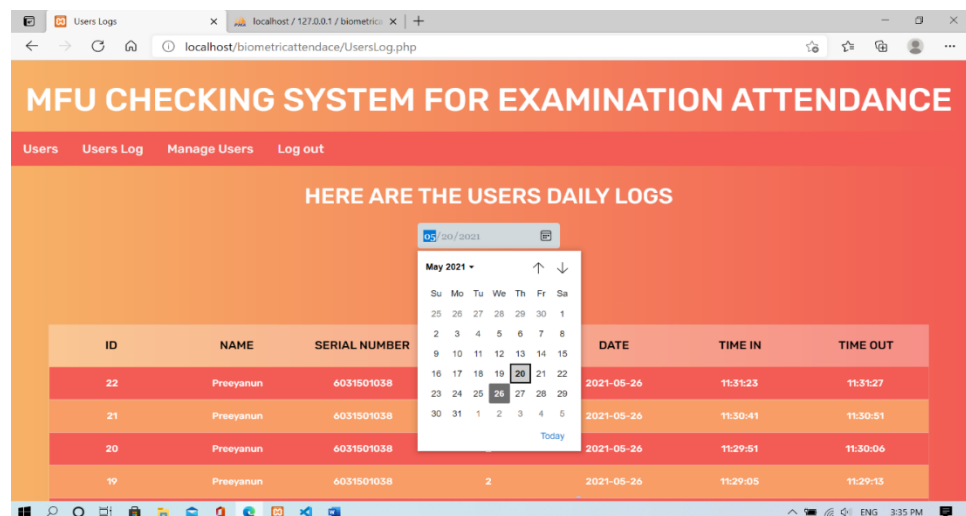
mm/dd/yyyy

Select Date

Export to Excel

ID	NAME	SERIAL NUMBER	FINGERPRINT ID	DATE	TIME IN	TIME OUT
22	Preeyanun	6031501038	2	2021-05-26	11:31:23	11:31:27
21	Preeyanun	6031501038	2	2021-05-26	11:30:41	11:30:51
20	Preeyanun	6031501038	2	2021-05-26	11:29:51	11:30:06
19	Preeyanun	6031501038	2	2021-05-26	11:29:05	11:29:13

Figure 4.20 Users login page (Admin) 1
Show Real-time fingerprint scanning list



MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE

Users Users Log Manage Users Log out

HERE ARE THE USERS DAILY LOGS

mm/dd/yyyy

May 2021

Su Mo Tu We Th Fr Sa

25 26 27 28 29 30 1

2 3 4 5 6 7 8

9 10 11 12 13 14 15

16 17 18 19 20 21 22

23 24 25 26 27 28 29

30 31 1 2 3 4 5

Today

ID	NAME	SERIAL NUMBER	FINGERPRINT ID	DATE	TIME IN	TIME OUT
22	Preeyanun	6031501038	2	2021-05-26	11:31:23	11:31:27
21	Preeyanun	6031501038	2	2021-05-26	11:30:41	11:30:51
20	Preeyanun	6031501038	2	2021-05-26	11:29:51	11:30:06
19	Preeyanun	6031501038	2	2021-05-26	11:29:05	11:29:13

Figure 4.21 Users login page (Admin) 2
When pressing to select the day that you want to view the historical data

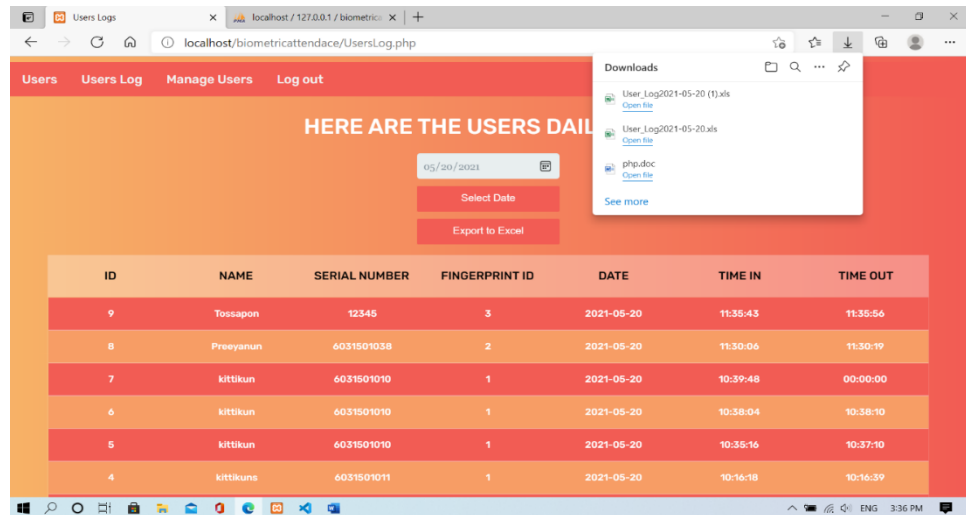


Figure 4.22 Users login page (Admin) 3
When clicking export to excel

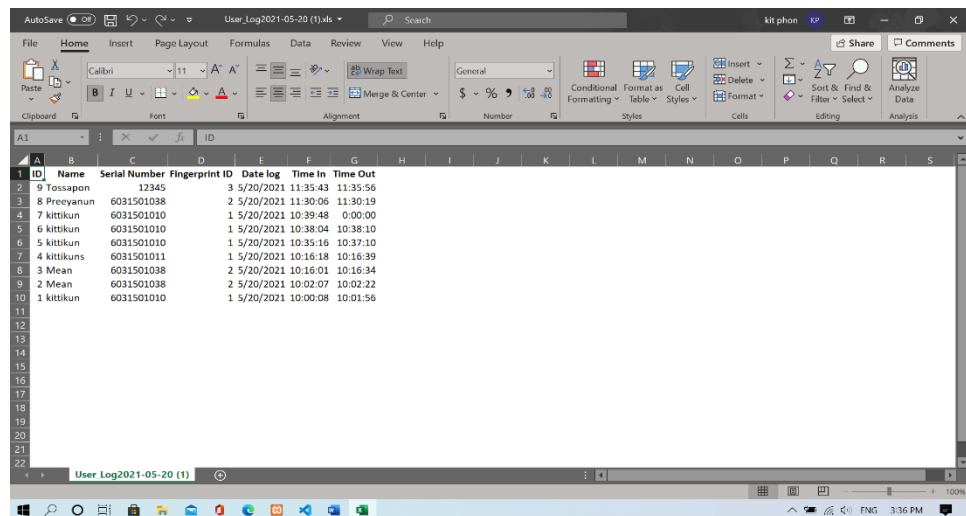


Figure 4.23 Excel File
Open the excel file after downloading. To view information

4.1.4.7 Login to the teacher webpage

The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/form_login.php'. The page title is 'Form Login'. It contains two input fields: 'ชื่อผู้ใช้' (Username) with the value 'kit' and 'รหัสผ่าน' (Password) with a masked value '***'. Below the fields are two green buttons: 'Login' and 'Reset'.

Figure 4.24 Teacher login page

4.1.4.8 A page showing real-time information when fingerprint scanning occurs to check the name of the student

The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/user_page.php'. The page has an orange background and a 'Log out' button in the top left. The main heading is 'HERE ARE THE USERS DAILY LOGS'. Below the heading are two buttons: 'Select Date' and 'Export to Excel'. A date picker shows 'mm/dd/yyyy'. Below these is a table with 7 columns: ID, NAME, SERIAL NUMBER, FINGERPRINT ID, DATE, TIME IN, and TIME OUT. The table contains 6 rows of data.

ID	NAME	SERIAL NUMBER	FINGERPRINT ID	DATE	TIME IN	TIME OUT
22	Preyasun	6031501038	2	2021-05-26	11:31:23	11:31:27
21	Preyasun	6031501038	2	2021-05-26	11:30:41	11:30:51
20	Preyasun	6031501038	2	2021-05-26	11:29:51	11:30:06
19	Preyasun	6031501038	2	2021-05-26	11:29:05	11:29:13
18	kitkan	6031501010	1	2021-05-26	11:28:35	00:00:00
17	kitkan	6031501010	1	2021-05-26	11:27:19	11:27:33

Figure 4.25 Users login page (Teacher)

4.2 Results of comparative case studies

Based on the fingerprint real-time attendance verification system with people who register fingerprints and don't register fingerprints. A fingerprint has been to compare the performance of the results clearly and accurate

4.2.1 Comparative testing

4.2.1.1 Registered completed result

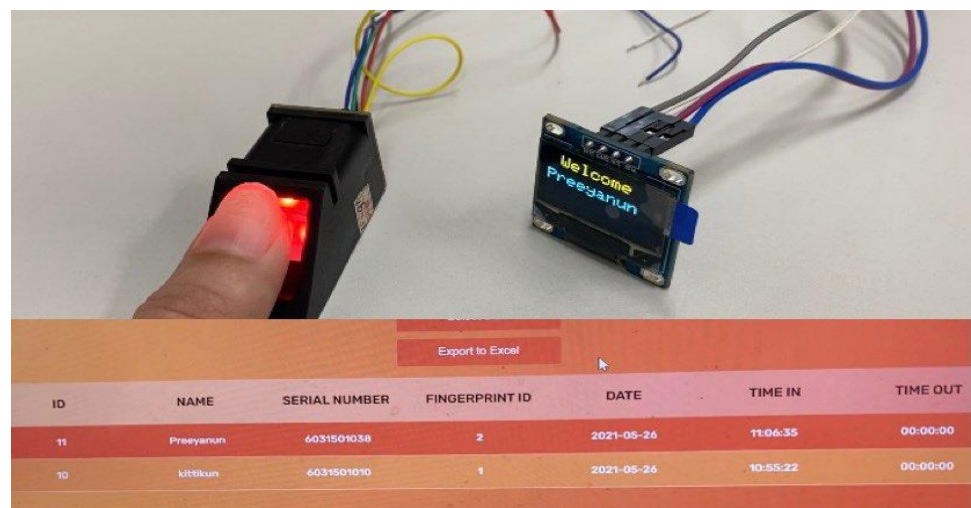


Figure 4.26 Registered completed result

The fingerprint that has been registered has been completed

4.2.1.2 Registered un-completed result

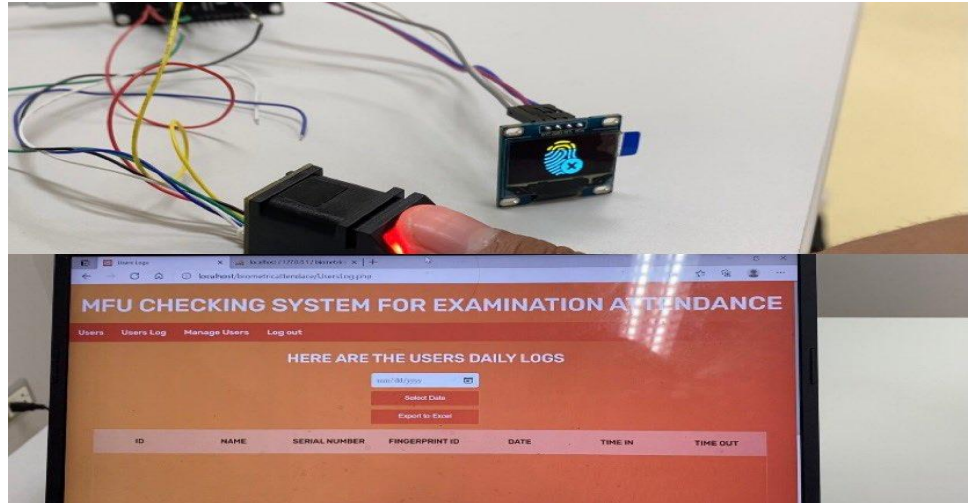


Figure 4.26 Registered uncompleted result
Fingerprint that has not been registered

CHAPTER 5

CONCLUSION

5.1 Conclusion

This project is a design of a real-time fingerprint examination attendance system consisting of a communicated fingerprint reader. Wirelessly with Wi-Fi signal, web application (web application) and MySQL database used to store information for examinations. In the design of the fingerprint reader, the Adafruit Optical Fingerprint Sensor FZ-1035 was selected as a fingerprint sensor. Along with various functions that have been written a set of instructions into the fingerprint verification machine, such as fingerprint authentication function, fingerprint registration function, some fingerprint deletion function, all fingerprints delete function. But in working, it uses the fingerprint authentication function mainly to verify the entrance examination. When the inspection is complete the end of the data is wirelessly transmitted to the database using the NodeMCU and the results of the examination are displayed in the web application. Web application and the database gets the results as designed can prove identity compare the data in the database. Display the results of the exam through the web application. And automatically record the exam results from the fingerprint scan of the registered person.

Also comparing the accuracy and clarity of fingerprint scanners. It was found that those who are not registered or do not have the information in the database will not be able to scanning entrance to the exam like those who have the information in the database. Therefore, the trial of this project meets the problem of exam fraud prevention. It is also more comfortable.

5.2 Problems

5.2.1. The use of fingerprints. When you use it for a long time, the fingerprint is not stable. Causing multiple scans to be scanned.

5.2.2. Connect to Wi-Fi in many forms. Which has to modify the code all the time when connecting to Wi-Fi in various places.

5.2.3. Coding management the connections between server XAMPP and client are difficult to manage.

5.3 Future work

5.3.1. Add more courses to divide the information so that users can view the information more systematically.

5.3.2. Web applications can be developed further, e.g. added chart summary page for easy visualization, the web can be opened on mobile for convenience.

5.3.3. It should be used in a variety of situations and increase the number of registered fingerprints.

5.3.4. Taken into the webserver and write down the name. To be able to use everywhere

5.3.5. Add more security to the database.

REFERENCES

- [1] Theory of Biometrics, <https://en.wikipedia.org/wiki/Biometrics>
- [2] Theory of Fingerprints, <https://en.wikipedia.org/wiki/Fingerprint>
- [3] Arches, Loops and Whorls of different types,
https://www.researchgate.net/figure/Basic-patterns-offingerprint_fig3_341989860
- [4] Technology and Tool, <https://robu.in/product/r307-optical-fingerprint-reader-module-sensor/>
- [5] NodeMCU ESP8266, <https://www.electroschematics.com/esp8266-datasheet/>
- [6] Apache XAMPP program, <https://community.spiceworks.com/products/57629-apache-xampp>
- [7] Visual Studio, <https://visualstudio.microsoft.com/>
- [8] Arduino IDE, <https://www.arduino.cc/en/software>
- [9] Fingerprint Sensor Based Self Bike Starter,
<https://how2electronics.com/fingerprint-sensor-based-self-bike-starter-using-arduino/>
- [10] Fingerprint Sensor Based Door Lock Security System,
<https://how2electronics.com/esp32-fingerprint-sensor-security-system/>
- [11] Fingerprint processing,
https://www.researchgate.net/publication/224646364_Design_of_an_Embedded_Fingerprint_Matcher_System
- [12] Website processing, <https://confidantech.com/web-design-methodology/>
- [13] Database processing, <https://sites.google.com/site/venusdtamturk/home/chapter-5-database-and-content-management>
- [14] Site Structure, <https://contentshifu.com/pillar/seo-site-structure>
- [15] Wire frame website, <https://www.thaibusinesssearch.com/marketing/wireframe/>

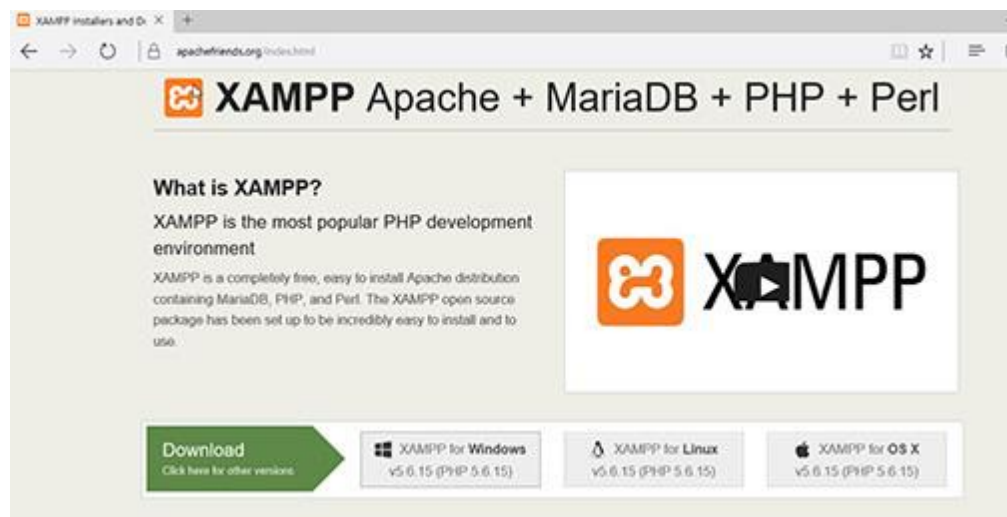
Appendix 1

User manual

1. Instruction install program

1.1 Install XAMPP to simulate a web server

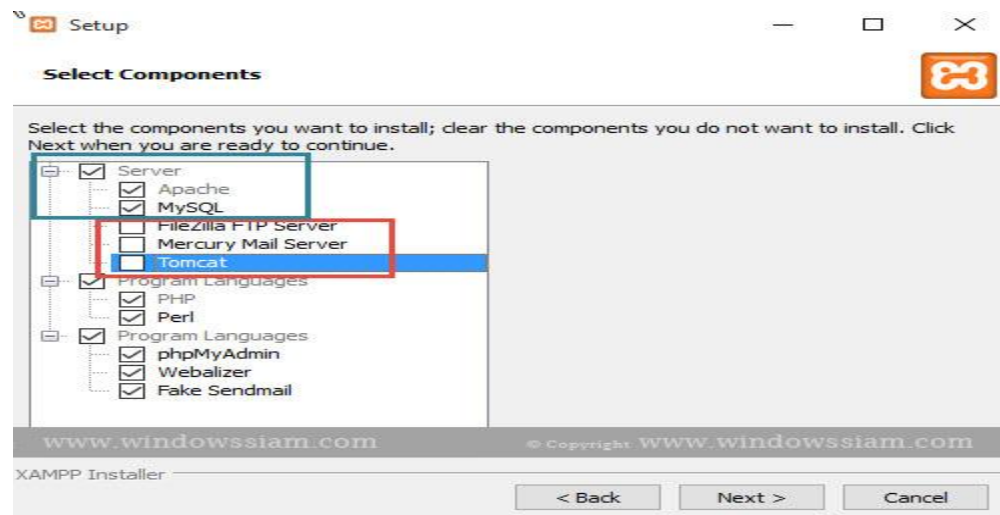
1.1.1 Download XAMPP for Windows.



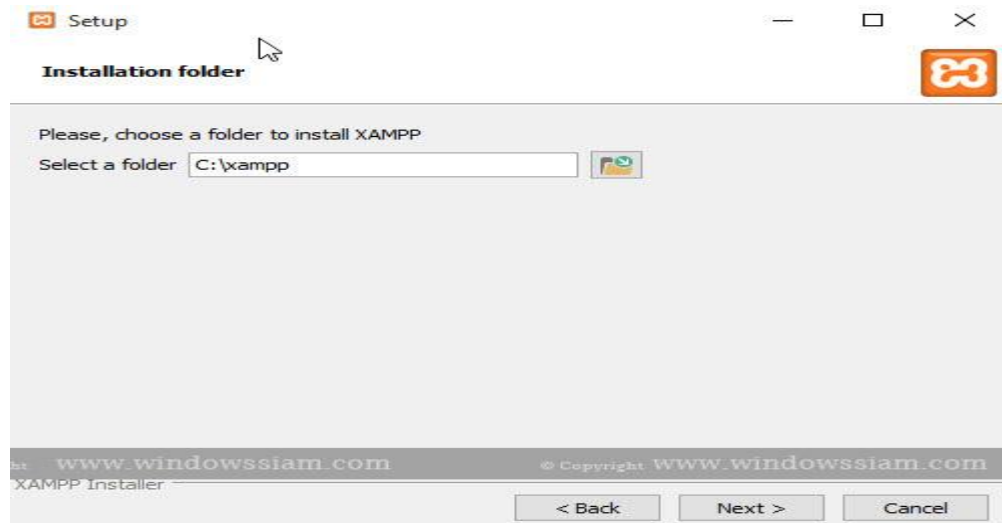
1.1.2 Click Next to install.



1.1.3 Select the components that we want.



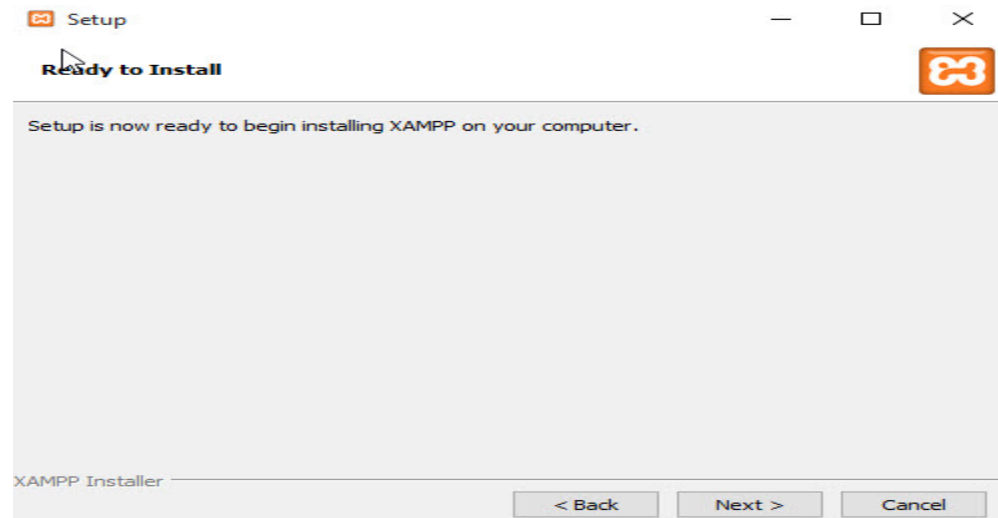
1.1.4 Select the path to install XAMPP.



1.1.5 Click Next.



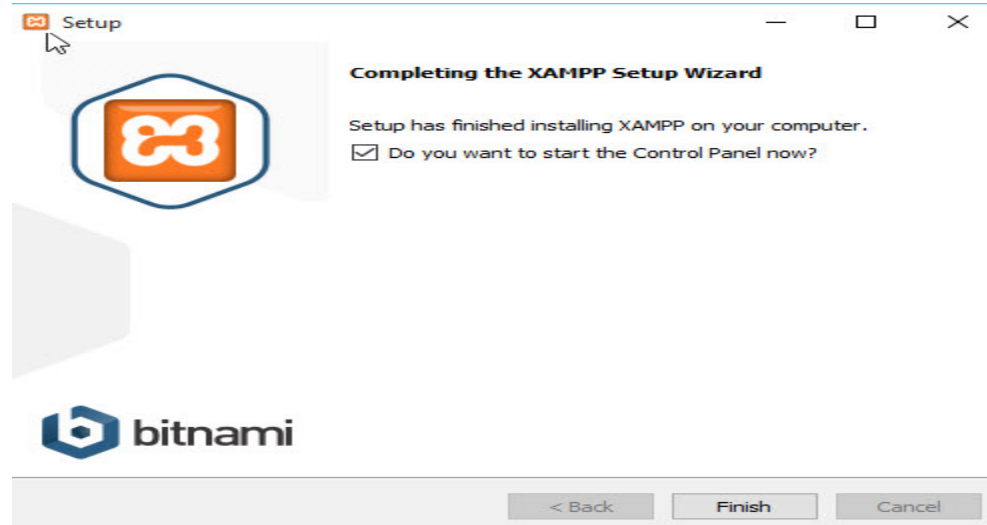
1.1.6 Click Next.



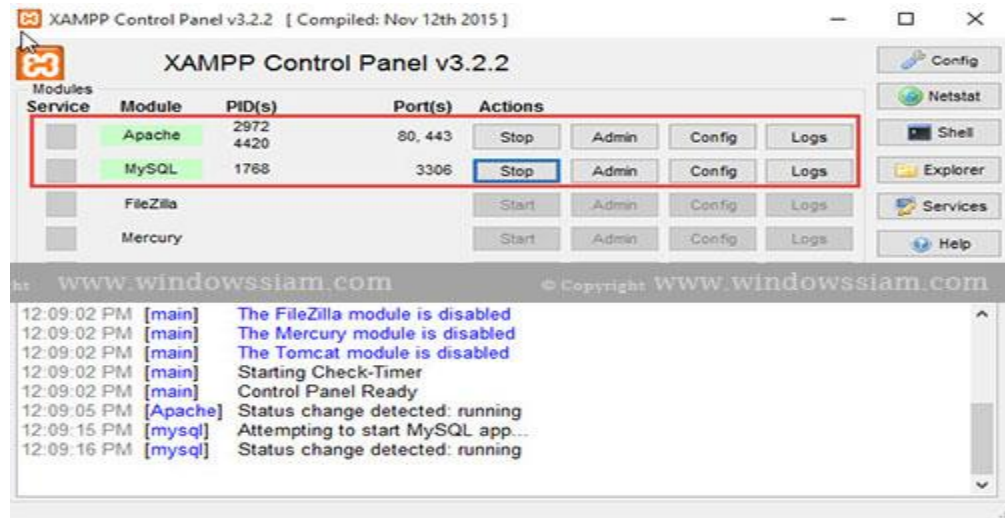
1.1.7 Wait for the installation.



1.1.8 Complete the installation and restart the computer once.



1.1.9 Then go to the XAMPP Control Panel and click on Start Apache and Mysql respectively.



1.1.10 Then try Phpmyadmin and localhost.

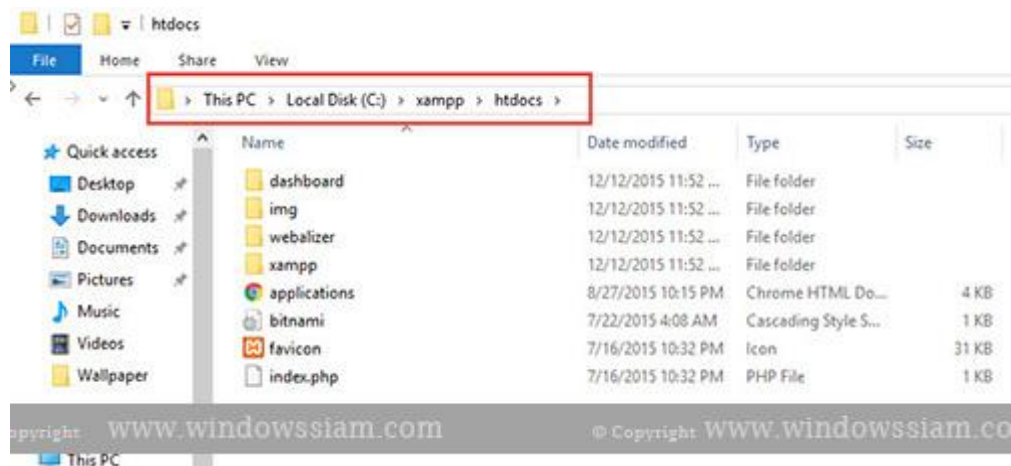
<http://localhost/phpmyadmin>

<http://localhost>



1.1.11 For Php Code in doing WebSite, take it to C: \ xampp \ htdocs \

Note: After we install xampp, we can add folders and files in htdocs.

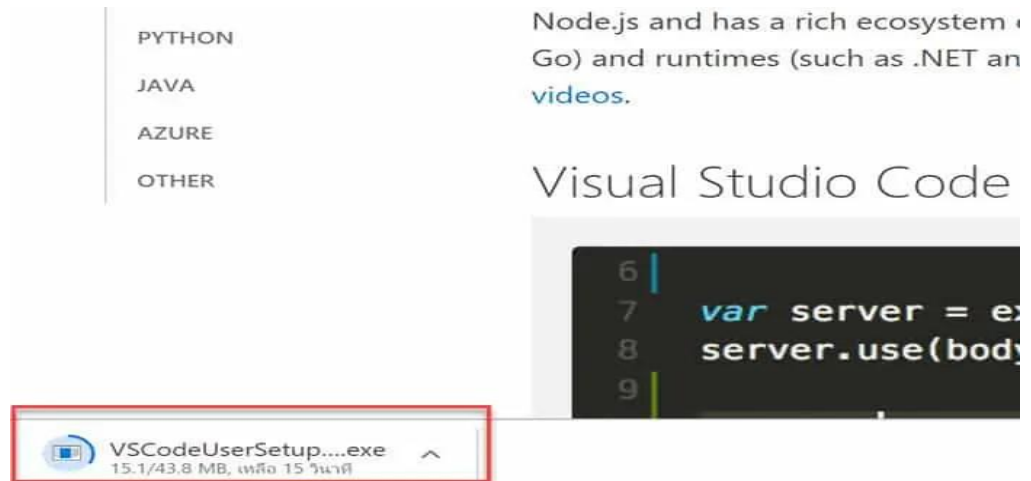


1.2 Install Visual Studio Code

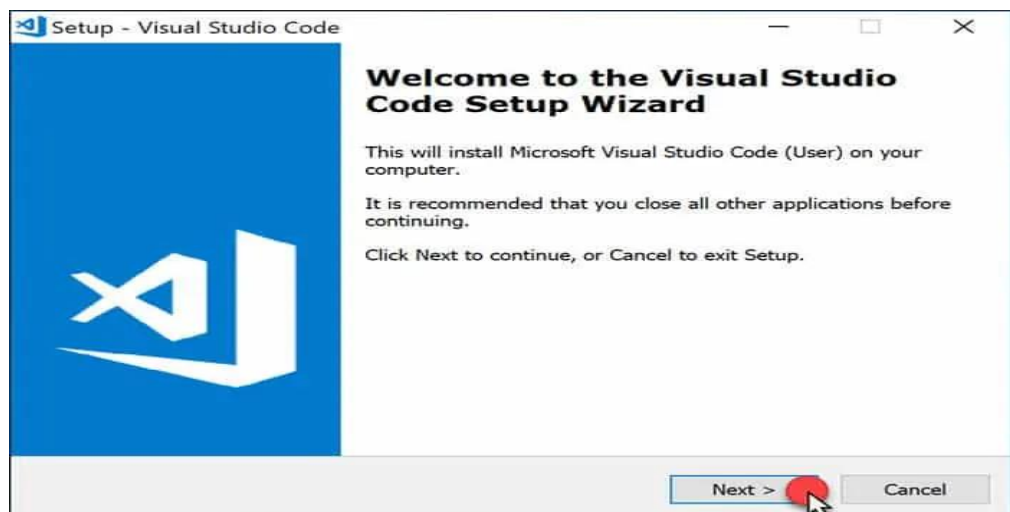
1.2.1 Download VS Code installer at <https://code.visualstudio.com>. When visiting the website, press the download button.



1.2.2 Wait for the installer to download. And when the download is complete, press to launch the installer.



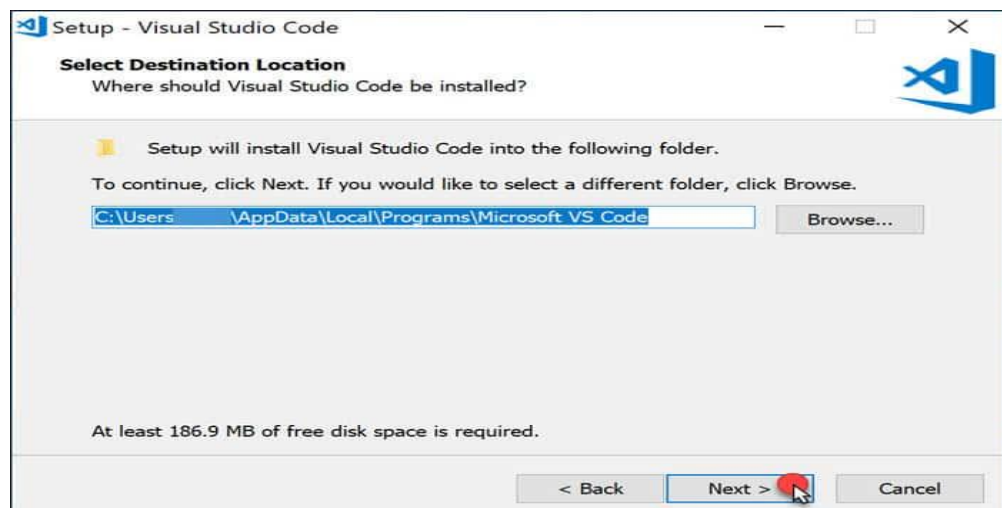
1.2.3 When the installer is launched, the Welcome to the Visual Studio Code Setup Wizard screen will appear, press the Next> button.



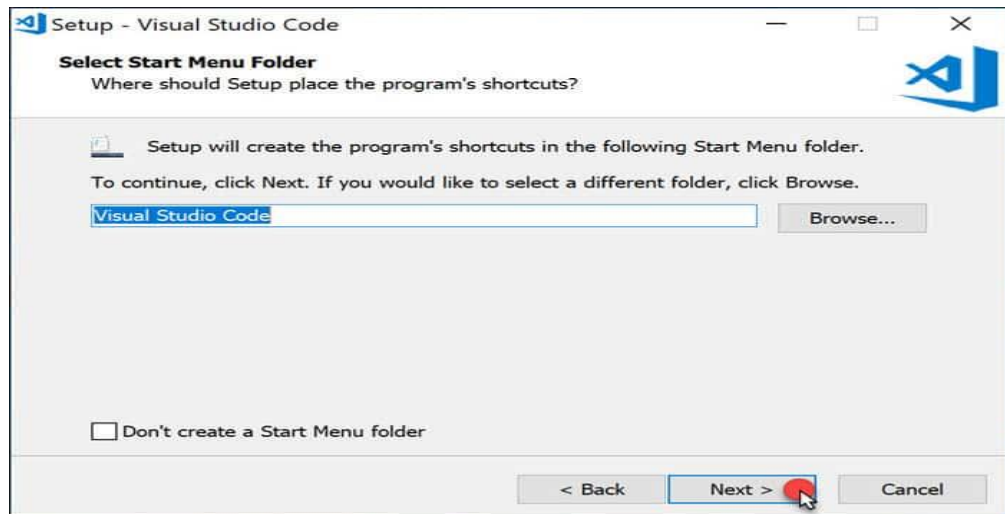
1.2.4 After that will display the License Agreement screen, select ☒ I accept the agreement and press Next>.



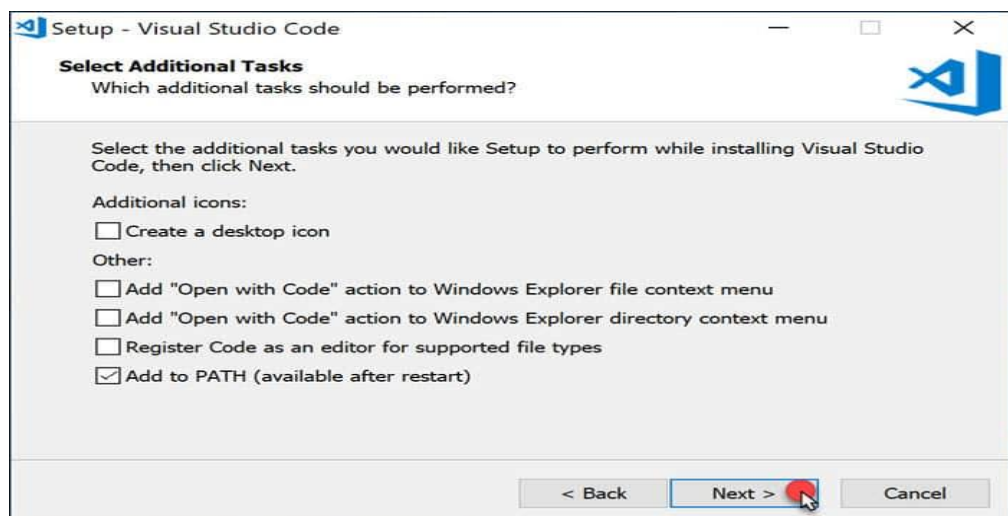
1.2.5 Then the screen Select Destination Location will be displayed, press Next> button.



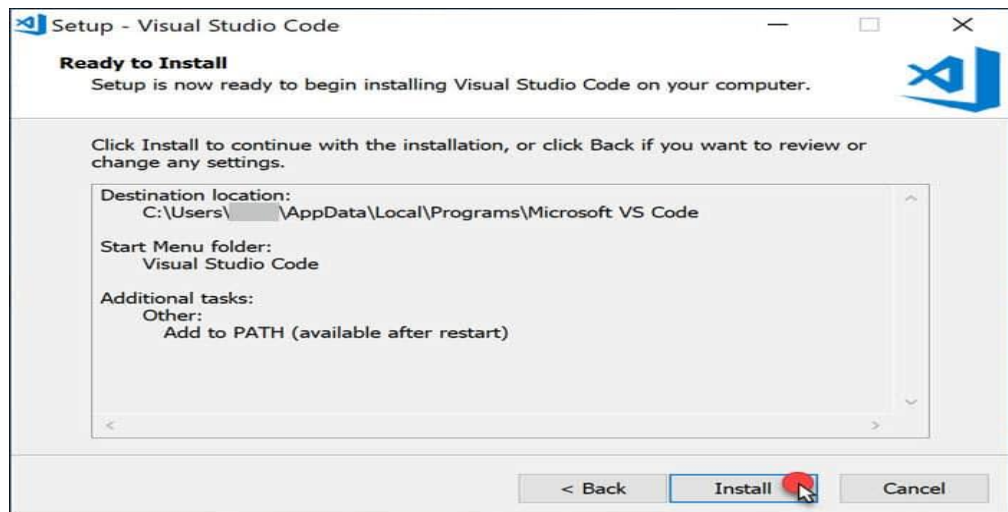
1.2.6 Then the Select Start Menu Folder screen will appear, press Next>.



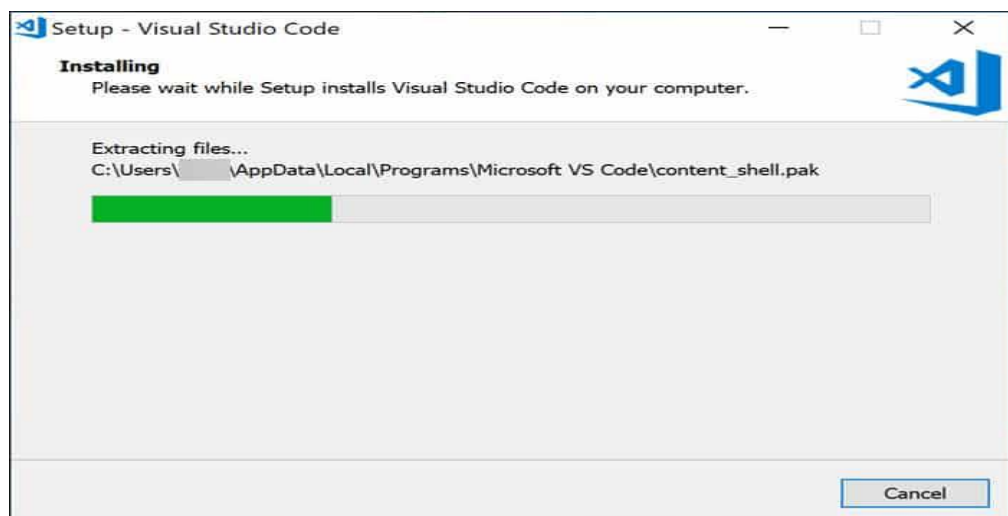
1.2.7 Then the Select Additional Tasks screen will appear, press the Next> button.



1.2.8 Then show the Ready to Install screen, press the Install button.



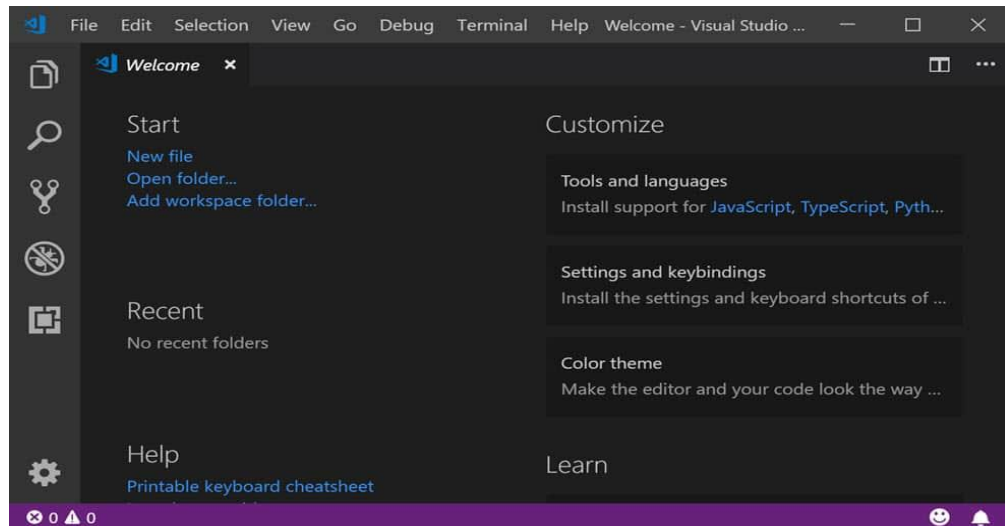
1.2.9 Then wait a moment for the program to install...



1.2.10 Once the installation is complete, it will display the Completing the Visual Studio Code Setup Wizard screen, press the Finish> button.



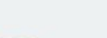
1.2.11 Then the Visual Studio Code program opens and completes the installation of Visual Studio Code.



1.3 Install the Arduino IDE with the ESP8266 and the sample program Library

1.3.1 Install Arduino IDE version 1.6.5 or later. The Arduino IDE can support Windows Mac OS X Linux 32bit and 64 bit. Download the installer from <https://www.arduino.cc/en/main/software>

Download the Arduino Software



ARDUINO 1.6.9

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. It runs on Windows, Mac OS X, and Linux. The environment is written in Java and based on Processing and other open-source software.

This software can be used with any Arduino board. Refer to the [Getting Started](#) page for installation instructions.

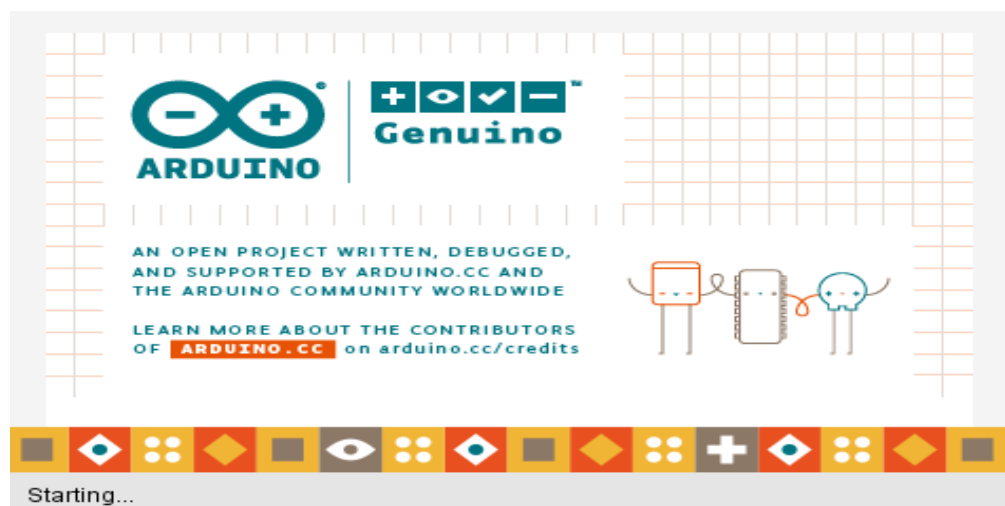
Windows Installer
Windows ZIP file for non admin install

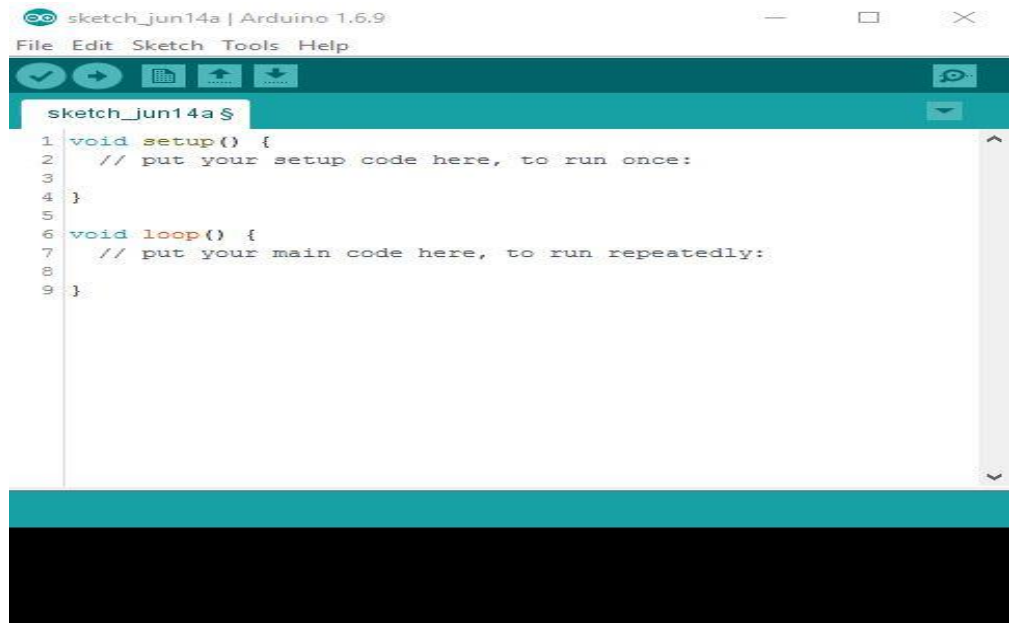
Mac OS X 10.7 Lion or newer

Linux 32 bits
Linux 64 bits
Linux ARM (experimental)

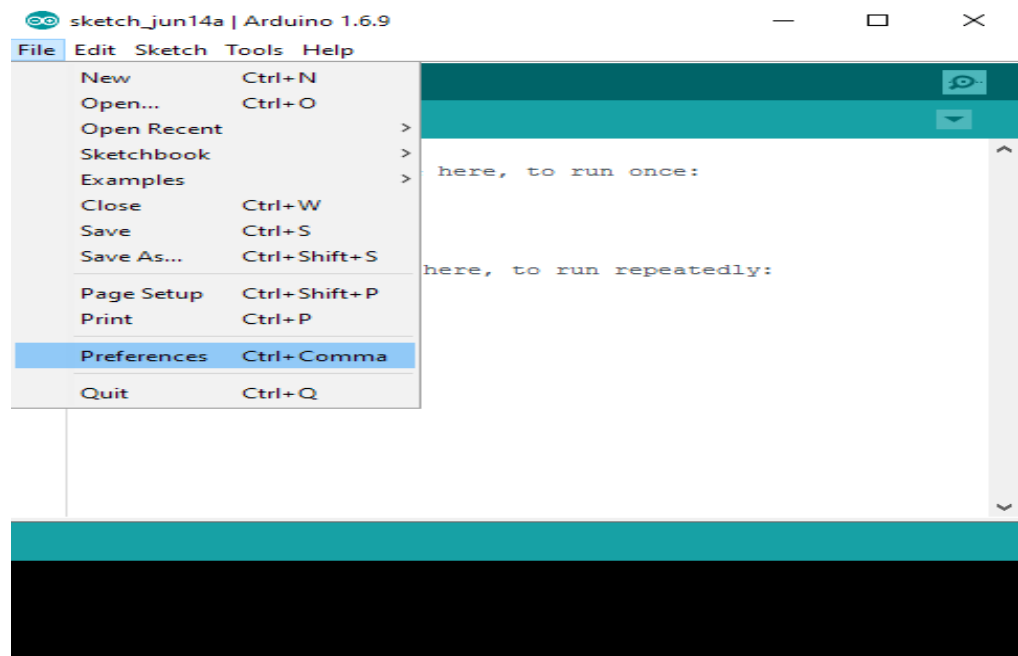
Release Notes
Source Code
Checksums

1.3.2 For installing the Arduino IDE on Windows, there are both options. Extract the files, use it (ZIP file for non admin install) or install like General program (Installer) If installed already, open Arduino IDE.

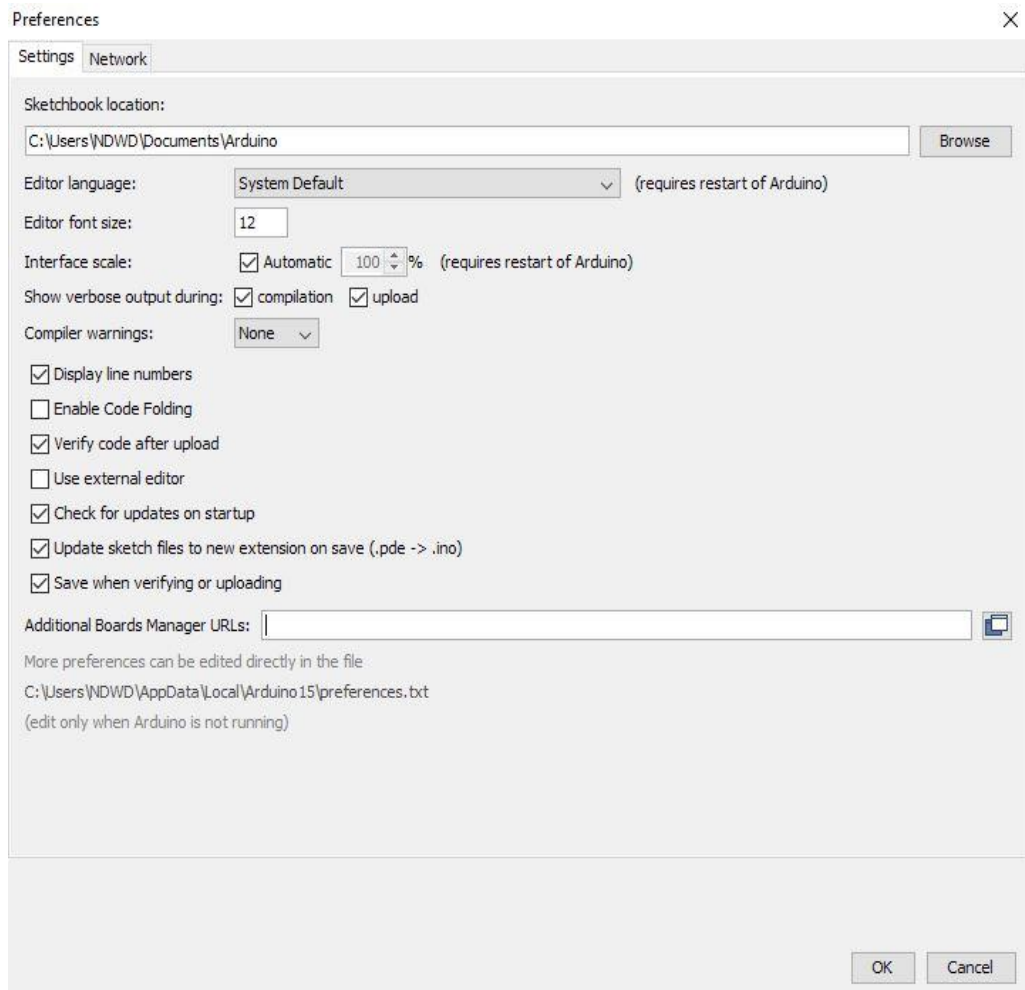




1.3.3 Install Broad ESP8266 onto Arduino IDE.
Select Menu to File >> Preferences



1.3.5 In the Preferences window, notice the Additional Board Manager URLs field:



1.3.6 As part of Additional Board Manager, this article uses the Community ESP8266 Boards Manager.

Available versions

Stable version updated Apr 18, 2016

Boards manager link: http://arduino.esp8266.com/stable/package_esp8266com_index.json

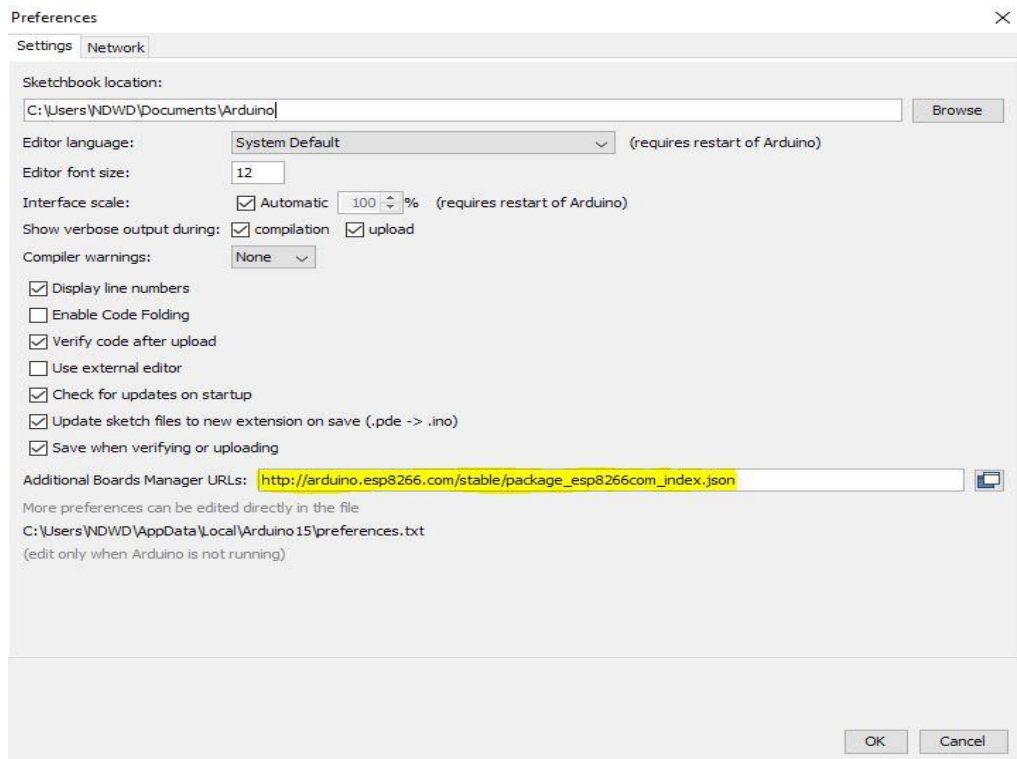
Documentation: <http://esp8266.github.io/Arduino/versions/2.2.0/>

Staging version updated May 11, 2016

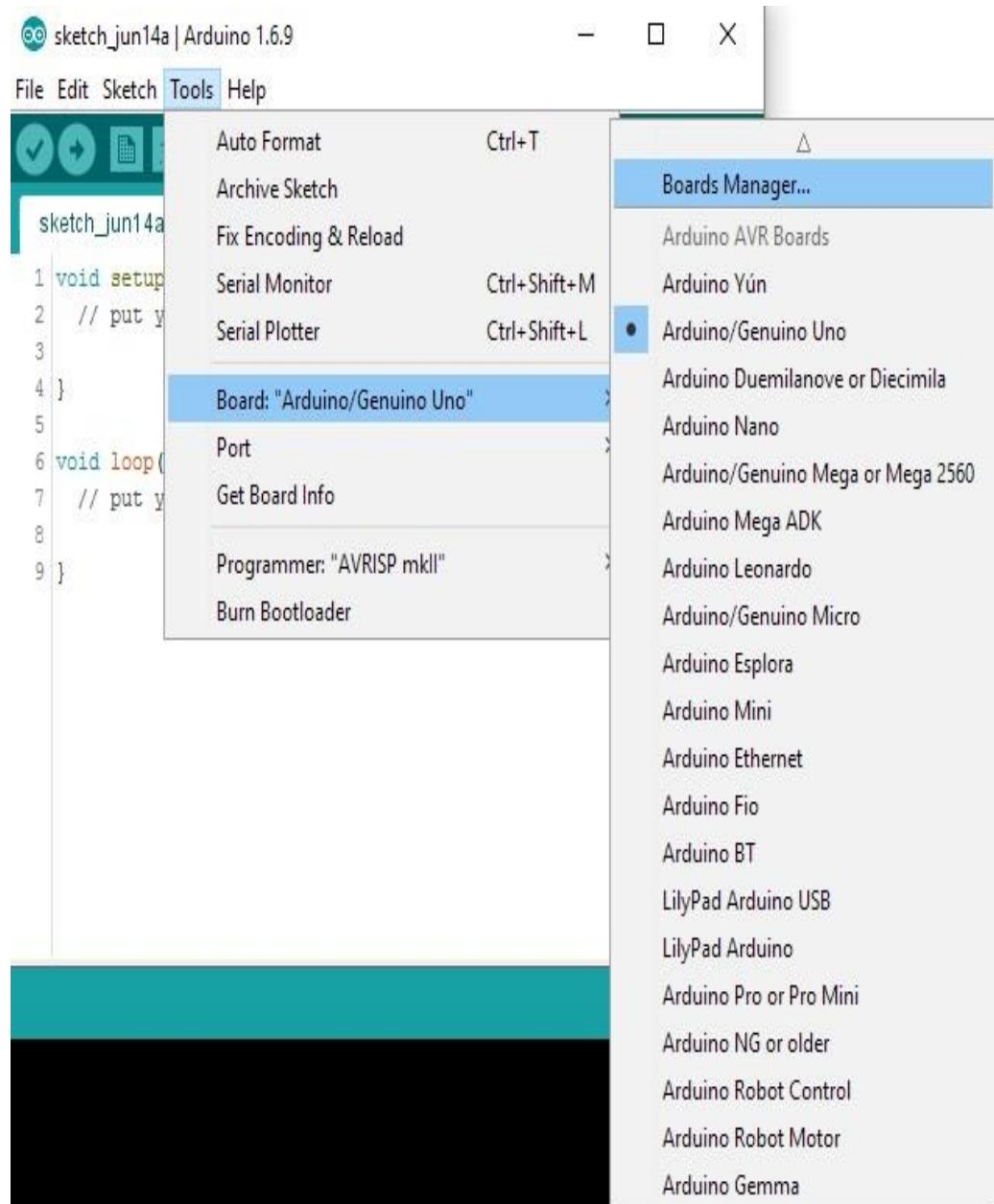
Boards manager link: http://arduino.esp8266.com/staging/package_esp8266com_index.json

Documentation: <http://esp8266.github.io/Arduino/versions/2.2.0-rc1/>

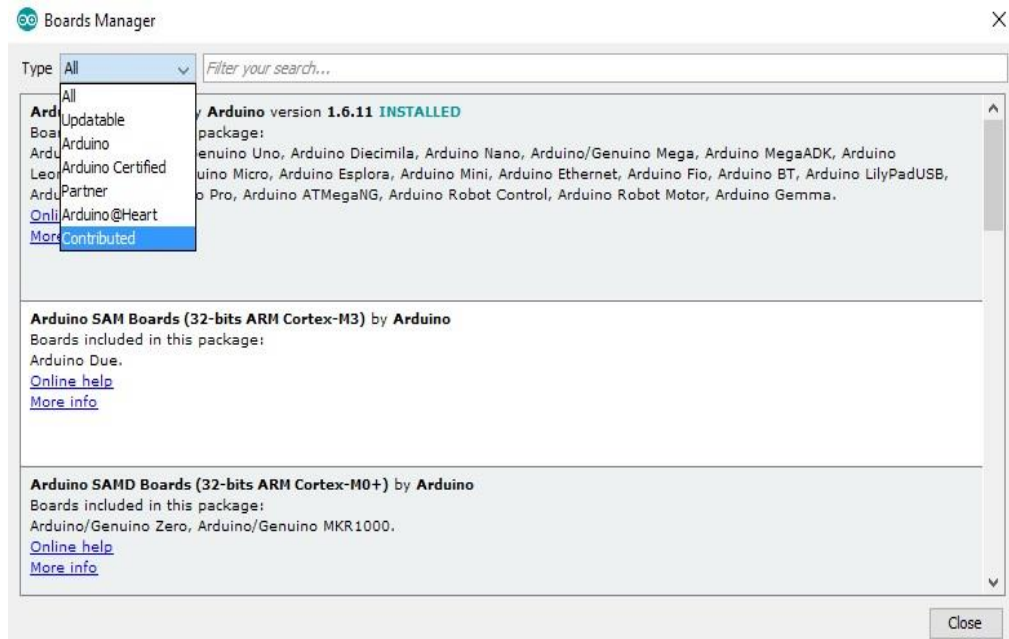
1.3.7 Put URL >> into Addition Board Manager URLs: as follows:
http://arduino.esp8266.com/stable/package_esp8266com_index.json Then
press OK.



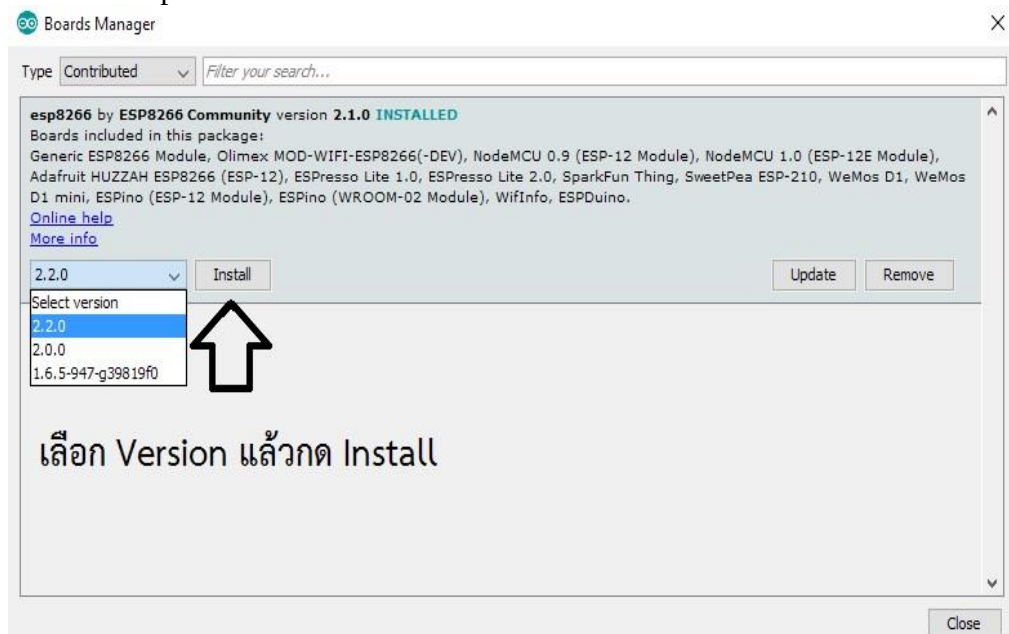
1.3.8 Go to Menu Tools >> Boar: “Arduino...” >> Board Manager...



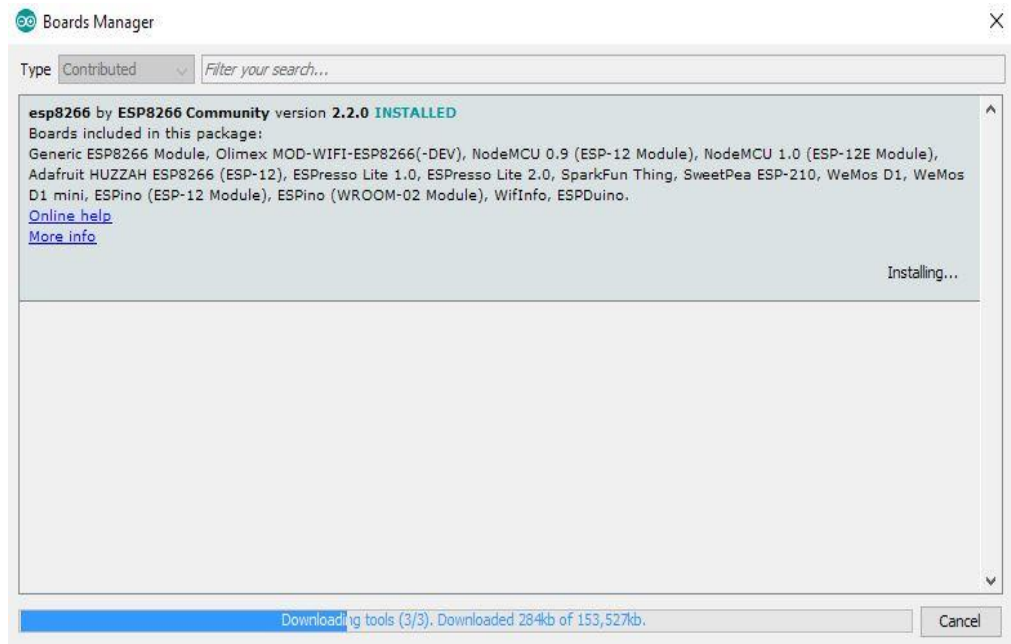
1.3.9 In the Boards Manager window, select Contributed type.



1.3.10 It will display Boards of ESP8266, select version 2.2.0 or later in future Boards and press Install.

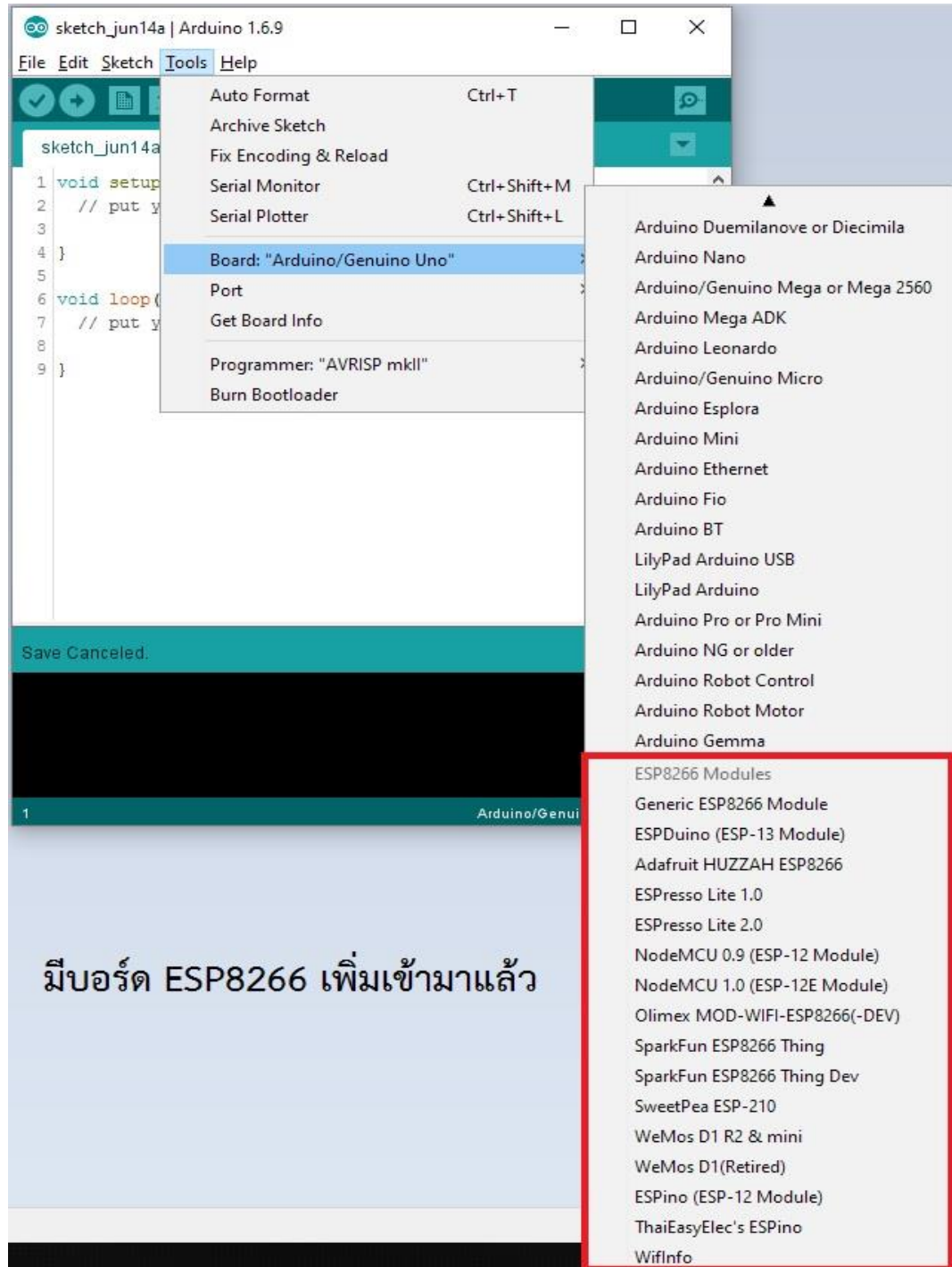


1.3.11 Then wait a few minutes, the program will load the ESP8266 Boards with a file size of about 150 MB and install it by yourself.



1.3.12 When we finished installing the ESP8266 board. Close the Arduino IDE program and open it up again.

When opening the Arduino IDE program again. Try to go to Menu Tools >> Board: "... .." You will find that there is a Menu for choosing to use ESP8266 with Arduino IDE.

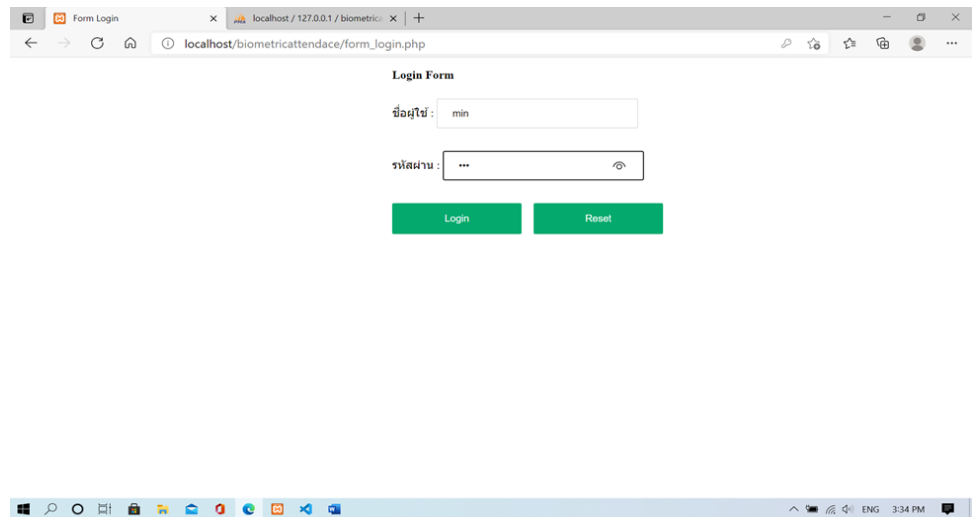


Appendix 2

System manual

2.1 Admin part

2.1.1 Login to the admin webpage



The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/form_login.php'. The page title is 'Form Login'. The login form contains two input fields: 'Username' with the value 'min' and 'Password' with three asterisks. Below the password field is a toggle icon for visibility. At the bottom of the form are two green buttons labeled 'Login' and 'Reset'.

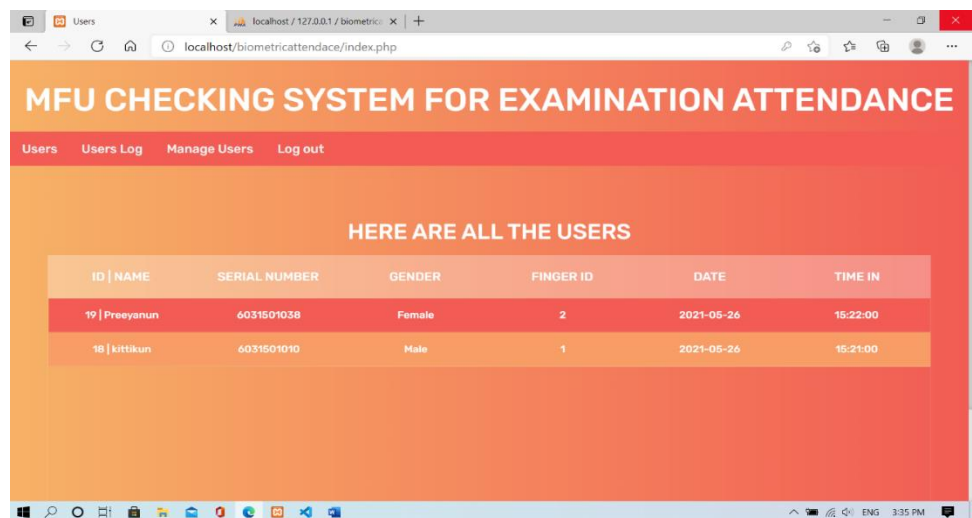
Form Login

Username: min

Password: ***

Login Reset

2.1.2 Page showing all user registered information



The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/index.php'. The page title is 'Users'. The page has a red header with the text 'MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE'. Below the header is a navigation bar with links: 'Users', 'Users Log', 'Manage Users', and 'Log out'. The main content area has a red background with the text 'HERE ARE ALL THE USERS'. Below this text is a table with the following data:

ID NAME	SERIAL NUMBER	GENDER	FINGER ID	DATE	TIME IN
19 Preeyanun	6031501038	Female	2	2021-05-26	15:22:00
18 kittikun	6031501010	Male	1	2021-05-26	15:21:00

2.1.3 Register fingerprint by specifying the fingerprint ID that you want to register

MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE

Users Users Log Manage Users Log out

ADD A NEW USER OR UPDATE HIS INFORMATION OR REMOVE HIM

The ID is ready to get a new Fingerprint

1 User Fingerprint ID:

Enter Fingerprint ID between 1 & 127:

User Fingerprint ID...

Add Fingerprint ID

2 User Info

User Name...

FINGER .ID	NAME	GENDER	S.NO	DATE	TIME IN
3	Name	Gender	0	2021-05-26	00:00:00
2	Preeyanun	Female	6031501038	2021-05-26	15:22:00
1	kittikun	Male	6031501010	2021-05-26	15:21:00

MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE

Users Users Log Manage Users Log out

ADD A NEW USER OR UPDATE HIS INFORMATION OR REMOVE HIM

User Fingerprint selected

1 User Fingerprint ID:

Enter Fingerprint ID between 1 & 127:

User Fingerprint ID...

Add Fingerprint ID

2 User Info

User Name...

Serial Number...

User Email...

3 Additional Info

Time In:

FINGER .ID	NAME	GENDER	S.NO	DATE	TIME IN
3	Name	Gender	0	2021-05-26	00:00:00
2	Preeyanun	Female	6031501038	2021-05-26	15:22:00
1	kittikun	Male	6031501010	2021-05-26	15:21:00

2.1.4 Add information by filling in the fields

Manage Users

localhost / 127.0.0.1 / biometric

localhost/biometricattendance/ManageUsers.php

1 User Fingerprint ID:

Enter Fingerprint ID between 1 & 127:

User Fingerprint ID...

Add Fingerprint ID

2 User Info

03 37 PM

Female Male

FINGER ID	NAME	GENDER	S.NO	DATE	TIME IN
3			0	2021-05-26	00:00:00
2	Preeyanun	Female	6031501038	2021-05-26	15:22:00
1	kittikun	Male	6031501010	2021-05-26	15:21:00

2.1.5 Press Add Fingerprint

Manage Users

localhost / 127.0.0.1 / biometric

localhost/biometricattendance/ManageUsers.php

Users Users Log Manage Users Log out

ADD A NEW USER OR UPDATE HIS INFORMATION OR REMOVE HIM

1 User Fingerprint ID:

Enter Fingerprint ID between 1 & 127:

User Fingerprint ID...

Add Fingerprint ID

2 User Info

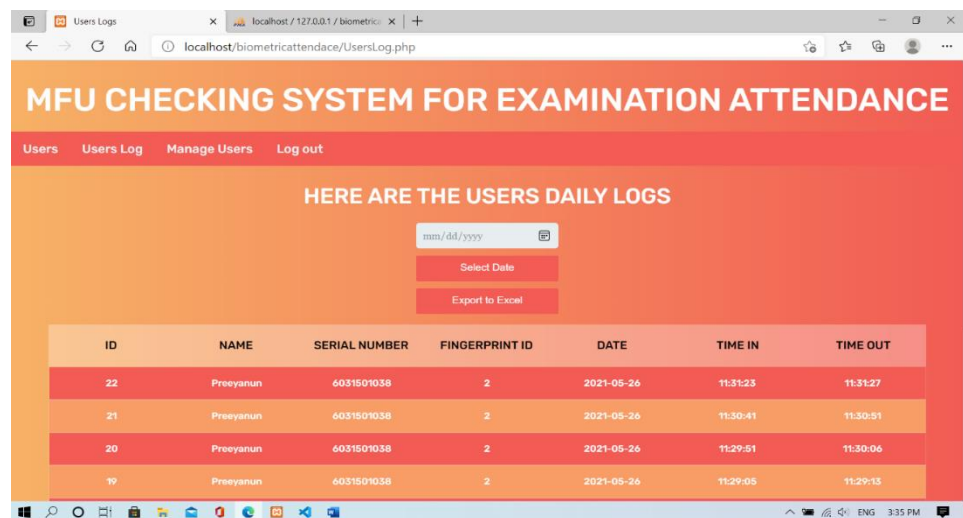
User Name...

Serial Number...

User Email...

FINGER ID	NAME	GENDER	S.NO	DATE	TIME IN
3	vivo	Male	60315010	2021-05-26	15:37:00
2	Preeyanun	Female	6031501038	2021-05-26	15:22:00
1	kittikun	Male	6031501010	2021-05-26	15:21:00

2.1.6 Showing real-time information when fingerprint scanning occurs. To check the name



MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE

Users Users Log Manage Users Log out

HERE ARE THE USERS DAILY LOGS

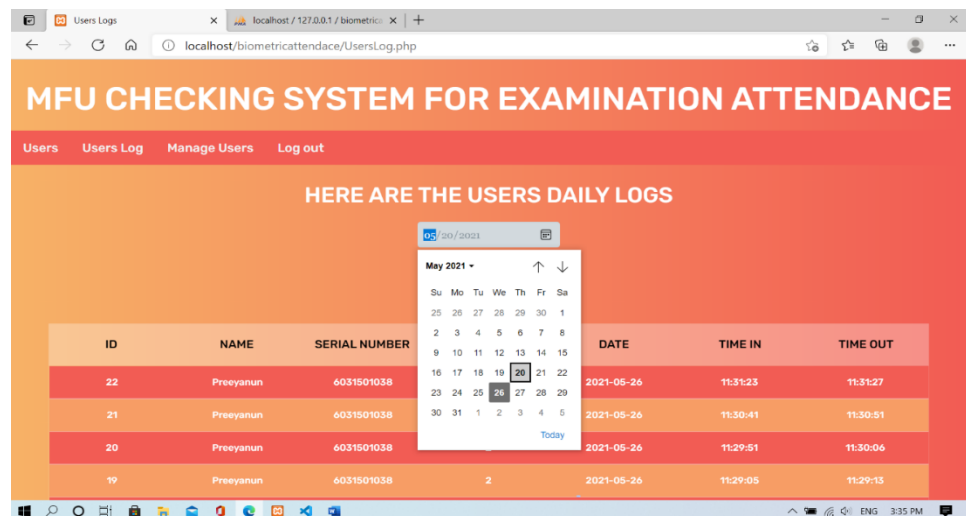
mm/dd/yyyy

Select Date

Export to Excel

ID	NAME	SERIAL NUMBER	FINGERPRINT ID	DATE	TIME IN	TIME OUT
22	Preeyanun	6031501038	2	2021-05-26	11:31:23	11:31:27
21	Preeyanun	6031501038	2	2021-05-26	11:30:41	11:30:51
20	Preeyanun	6031501038	2	2021-05-26	11:29:51	11:30:06
19	Preeyanun	6031501038	2	2021-05-26	11:29:05	11:29:13

2.1.7 Select the day that you want to view the historical data



MFU CHECKING SYSTEM FOR EXAMINATION ATTENDANCE

Users Users Log Manage Users Log out

HERE ARE THE USERS DAILY LOGS

05/26/2021

May 2021

Su Mo Tu We Th Fr Sa

25 26 27 28 29 30 1

2 3 4 5 6 7 8

9 10 11 12 13 14 15

16 17 18 19 20 21 22

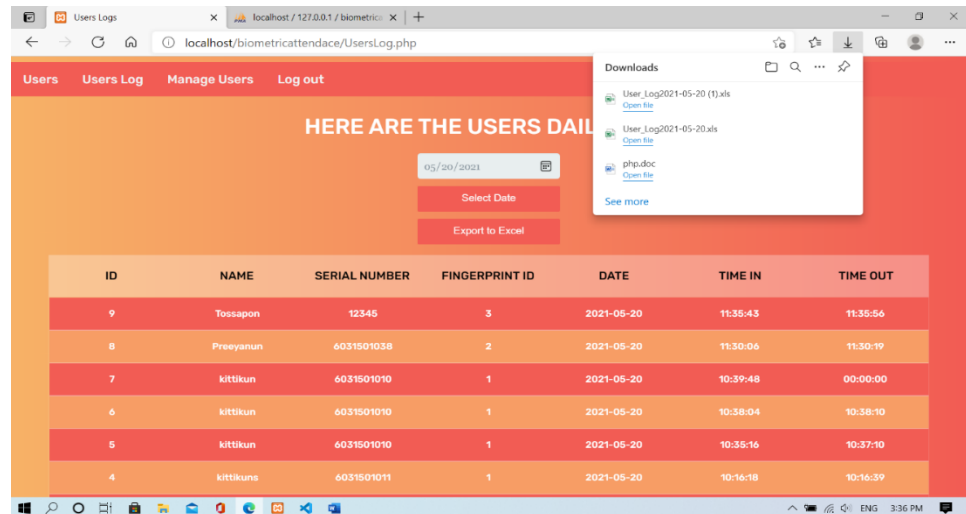
23 24 25 26 27 28 29

30 31 1 2 3 4 5

Today

ID	NAME	SERIAL NUMBER	DATE	TIME IN	TIME OUT
22	Preeyanun	6031501038	2021-05-26	11:31:23	11:31:27
21	Preeyanun	6031501038	2021-05-26	11:30:41	11:30:51
20	Preeyanun	6031501038	2021-05-26	11:29:51	11:30:06
19	Preeyanun	6031501038	2021-05-26	11:29:05	11:29:13

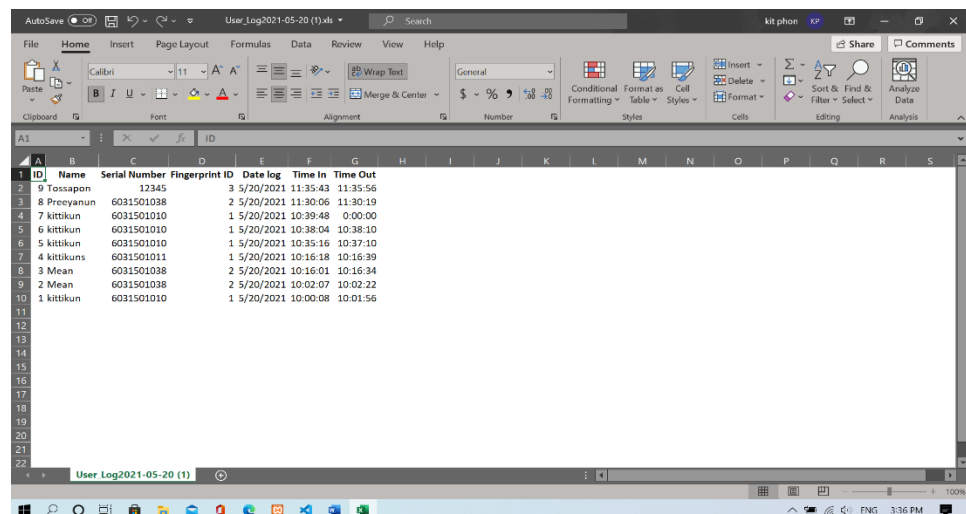
2.1.8 Clicking export to excel



The screenshot shows a web application interface for viewing user logs. The page has a red header with navigation links: "Users", "Users Log", "Manage Users", and "Log out". Below the header, there's a section titled "HERE ARE THE USERS DAILY" with a date selector set to "05/20/2021". There are two buttons: "Select Date" and "Export to Excel". A table of user logs is displayed below the buttons. A "Downloads" menu is open on the right side of the browser, showing two files: "User_Log2021-05-20 (1).xls" and "User_Log2021-05-20.xls", both with "Open file" links. The table contains the following data:

ID	NAME	SERIAL NUMBER	FINGERPRINT ID	DATE	TIME IN	TIME OUT
9	Tossapon	12345	3	2021-05-20	11:35:43	11:35:56
8	Preeyanun	6031501038	2	2021-05-20	11:30:06	11:30:19
7	kittikun	6031501010	1	2021-05-20	10:39:48	00:00:00
6	kittikun	6031501010	1	2021-05-20	10:38:04	10:38:10
5	kittikun	6031501010	1	2021-05-20	10:35:16	10:37:10
4	kittikuns	6031501011	1	2021-05-20	10:16:18	10:16:39

2.1.9 Open the excel file after downloading to view information

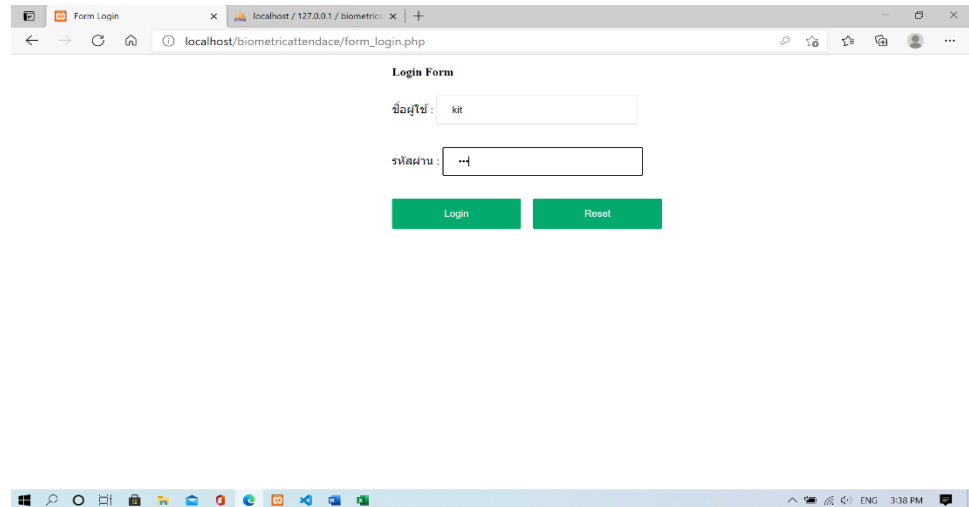


The screenshot shows an Excel spreadsheet titled "User_Log2021-05-20 (1).xls". The spreadsheet contains the following data:

ID	Name	Serial Number	Fingerprint ID	Date log	Time In	Time Out
9	Tossapon	12345	3	5/20/2021	11:35:43	11:35:56
8	Preeyanun	6031501038	2	5/20/2021	11:30:06	11:30:19
7	kittikun	6031501010	1	5/20/2021	10:39:48	00:00:00
6	kittikun	6031501010	1	5/20/2021	10:38:04	10:38:10
5	kittikun	6031501010	1	5/20/2021	10:35:16	10:37:10
4	kittikuns	6031501011	1	5/20/2021	10:16:18	10:16:39
3	Mean	6031501038	2	5/20/2021	10:16:01	10:16:34
2	Mean	6031501038	2	5/20/2021	10:02:07	10:02:22
1	kittikun	6031501010	1	5/20/2021	10:00:08	10:01:56

2.2 Teacher part

2.2.1 Login to the teacher webpage



The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/form_login.php'. The page title is 'Form Login'. It contains a login form with two input fields: 'ชื่อผู้ใช้:' (Username) with the value 'kit' and 'รหัสผ่าน:' (Password) with the value '123'. Below the fields are two green buttons labeled 'Login' and 'Reset'.

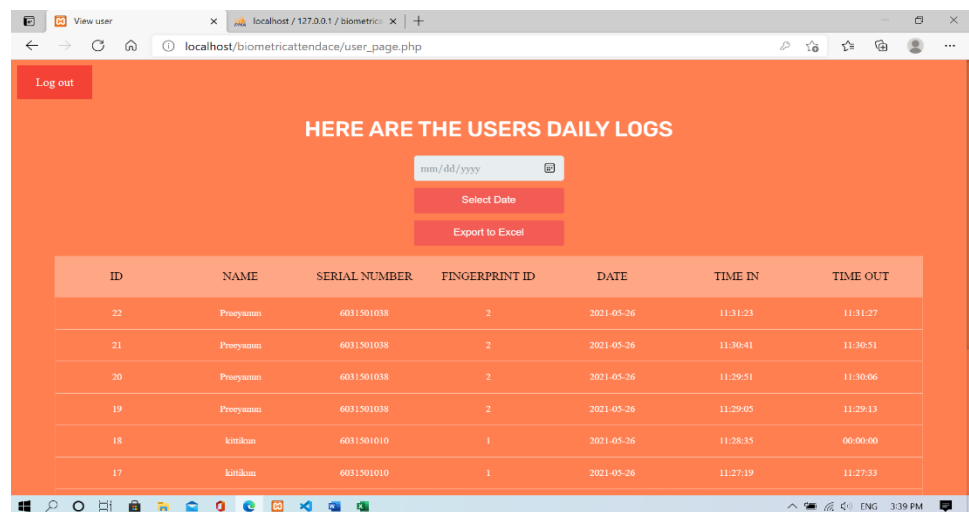
Form Login

ชื่อผู้ใช้: kit

รหัสผ่าน: 123

Login Reset

2.2.2 A page showing real-time information when fingerprint scanning occurs to check the name of the student



The screenshot shows a web browser window with the address bar displaying 'localhost/biometricattendance/user_page.php'. The page title is 'View user'. It features a red 'Log out' button in the top left corner. The main heading is 'HERE ARE THE USERS DAILY LOGS'. Below the heading is a date selection interface with a text input 'mm/dd/yyyy', a 'Select Date' button, and an 'Export to Excel' button. A table displays the daily logs with columns: ID, NAME, SERIAL NUMBER, FINGERPRINT ID, DATE, TIME IN, and TIME OUT. The table contains six rows of data.

Log out

HERE ARE THE USERS DAILY LOGS

mm/dd/yyyy

Select Date

Export to Excel

ID	NAME	SERIAL NUMBER	FINGERPRINT ID	DATE	TIME IN	TIME OUT
22	Prayasan	6031501038	2	2021-05-26	11:31:23	11:31:27
21	Prayasan	6031501038	2	2021-05-26	11:30:41	11:30:51
20	Prayasan	6031501038	2	2021-05-26	11:29:51	11:30:06
19	Prayasan	6031501038	2	2021-05-26	11:29:05	11:29:13
18	kitikae	6031501010	1	2021-05-26	11:28:35	00:00:00
17	kitikae	6031501010	1	2021-05-26	11:27:19	11:27:33